A COURSE ON INDUCTION FURNACE

&

TROUBLE SHOOTING

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INTRODUCTION

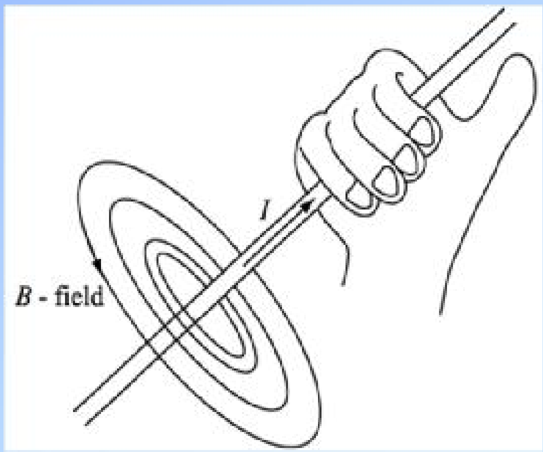
Electromagnetism

And

Principle Of Electromagnetic Induction

ELECTROMAGNETISM

- Current carrying conductors are associated with magnetic field surrounding it in concentric circle.
- Stronger the current , stronger is the magnetic field produce.
- Reverse is also true , i.e a wire moving in magnetic field has electrical current generated in it.
- Direction of magnetic field and flow of current is given by RIGHT HAND RULE.



ELECTROMAGNETIC INDUCTION

- The process of generating current in a conductor by placing the conductor in a changing magnetic field is called INDUCTION.
- Name induction because there is no physical connection between the conductor and the magnet.
- For Electromagnetic Induction to take place, the conductor which is often a piece of wire, must be perpendicular to the magnetic lines of force and should cut the lines of magnetic field.
- Principle of electromagnetic induction is stated by Michael Faraday in 1831.

FARADAY LAW OF ELECTROMAGNETIC INDUCTION

- **Faraday's First Law :**

  Whenever the magnetic flux linked with a circuit changes, an e.m.f. is always induced in it.

- **Faraday's Second Law** :

  The magnitude of the induced e.m.f. is equal to the rate of change of flux linkages.

  (Induced emf = N $d\phi/dt$) , N= No. of Turn & ϕ= flux.

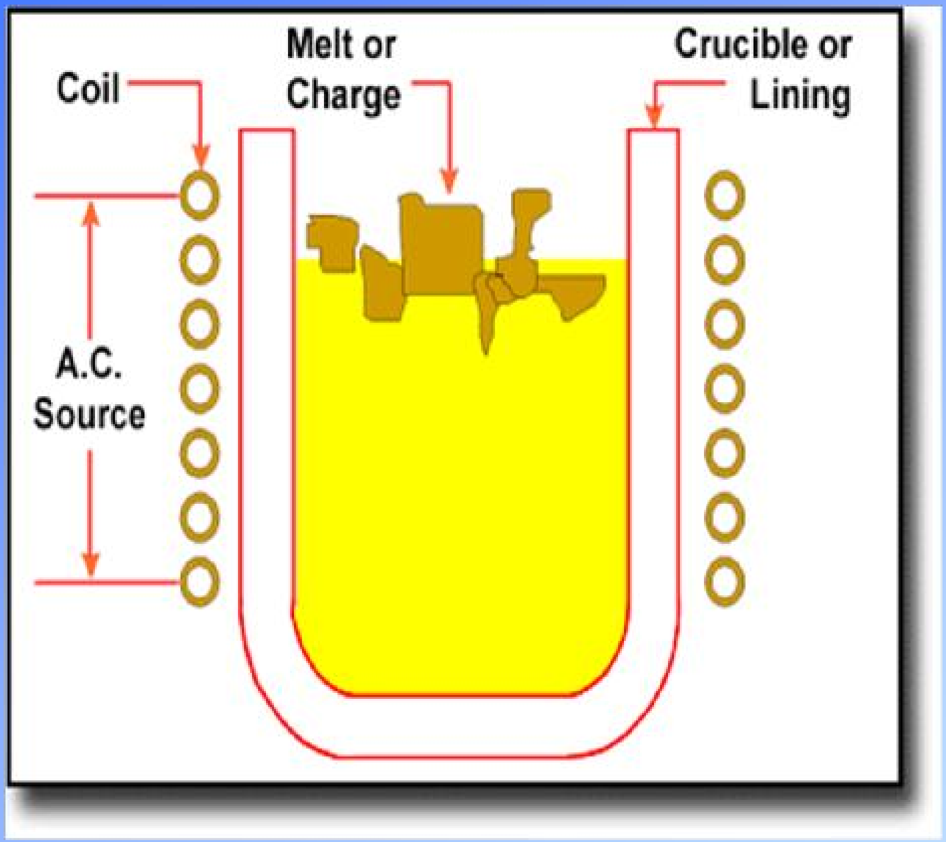
INDUCTION HEATING/MELTING

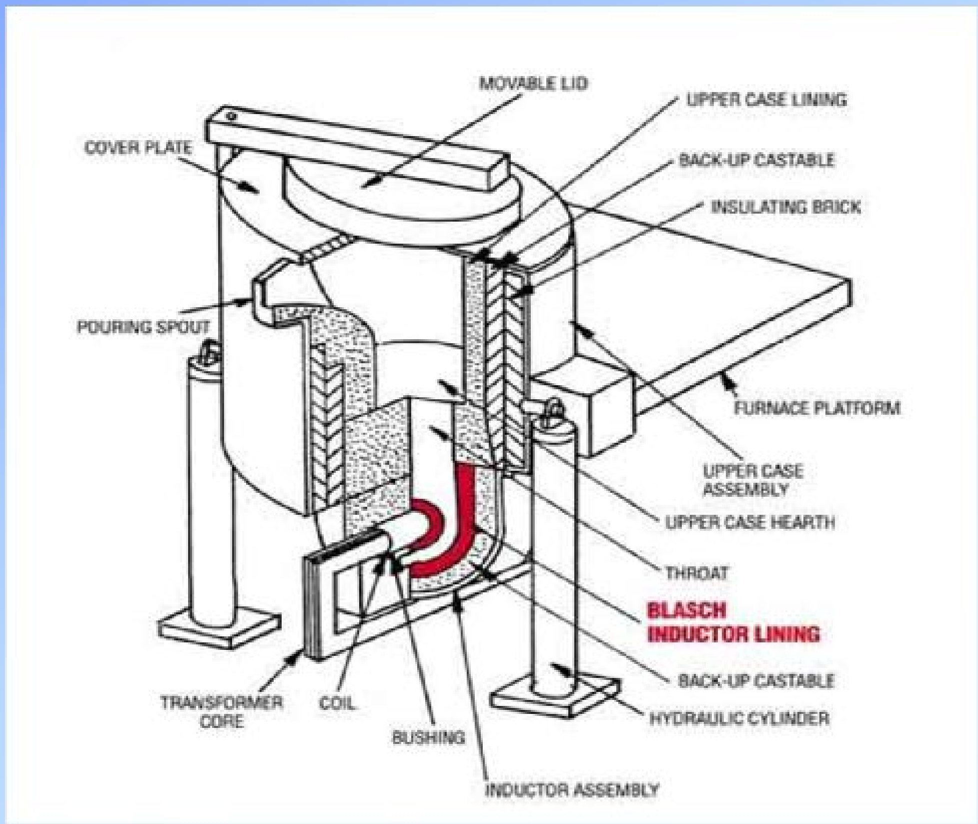
- If an AC current is fed through a coil, the electromagnetic field that is produced is constantly changing. This changing magnetic field induces electrical current (AC) in another circuit that is held close .
- This induced secondary current can be utilized to generate heat in the circuit of the second conductor by exploring the electrical losses due to the current. These losses are due to Hysterises and Eddy current.

INDUCTION MELTING FURNACE

- Coreless Furnace

Channel Furnace



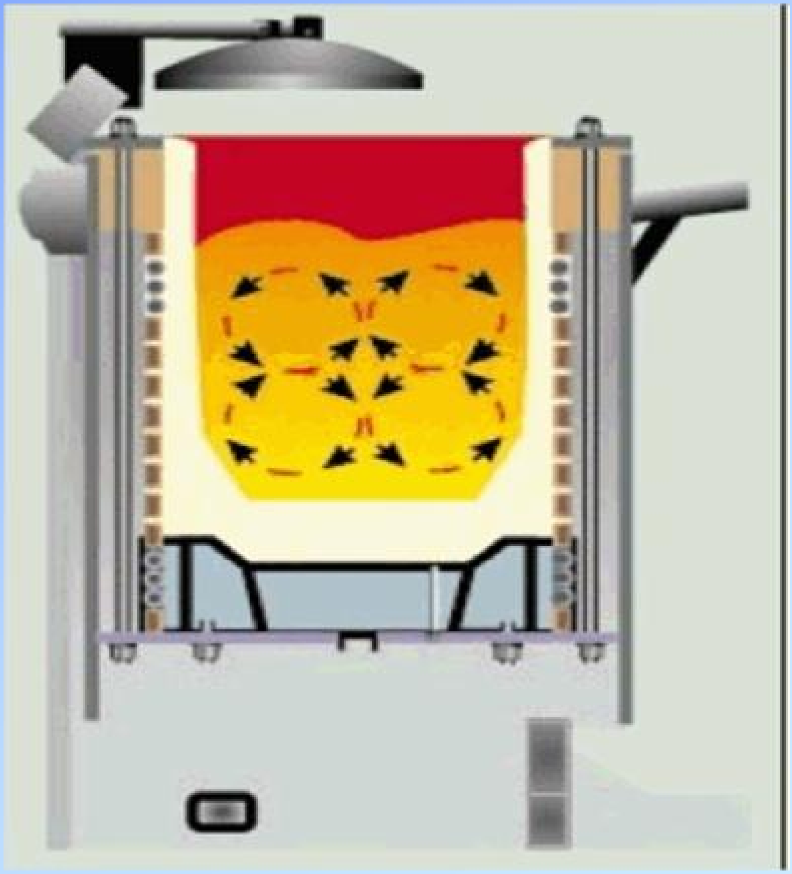


CORELESS INDUCTION FURNACE

- In this type of furnace the coil is hollow and metal scrap is charge in it for melting. Core is not present for flux linkage between primary and secondary and hence the name.
- The heart of the coreless induction furnace is the coil, which consists of a hollow section of heavy duty, high conductivity copper tubing which is wound into a helical coil.
- Coil shape is contained within a steel shell and magnetic shielding is used to prevent heating of the supporting shell.
- To protect it from overheating, the coil is water-cooled, the water being recirculated and cooled in a cooling tower. The shell is supported on trunnions on which the furnace tilts to facilitate pouring.
- The solid state generator panel converts the voltage and frequency of main supply, to that required for electrical melting.

CORELESS INDUCTION FURNACE …cont

- When the charge material is molten, the interaction of the magnetic field and the electrical currents flowing in the induction coil produce a stirring action within the molten metal.
- The degree of stirring action is influenced by the power and frequency applied as well as the size and shape of the coil and the density and viscosity of the molten metal.
- The stirring action within the bath is important as it helps with mixing of alloys and melting of turnings as well as homogenizing of temperature throughout the furnace.



Stirring Action

CORELESS INDUCTION FURNACE …cont

- Excessive stirring can increase gas pick up, lining wear and oxidation of alloys.
- The crucible is formed by ramming a granular refractory between the coil and a hollow internal former (M.S cylinder) which is melted away with the first heat leaving a sintered lining.
- Successive coil turns are electrically insulated from each other using several layers of high voltage , high temperature abrasive resistant polymer coating.
- Spacing and total height is maintained by bolting each turns of coil with fiberglass channel, which prevent coil deformation.
- The coreless induction furnace is commonly used to melt all grades of steels and irons as well as many non-ferrous alloys.
- The furnace is ideal for re-melting and alloying because of the high degree of control over temperature and chemistry while the induction current provides good circulation of the melt.

ET- INDUCTION MELTING FURNACE

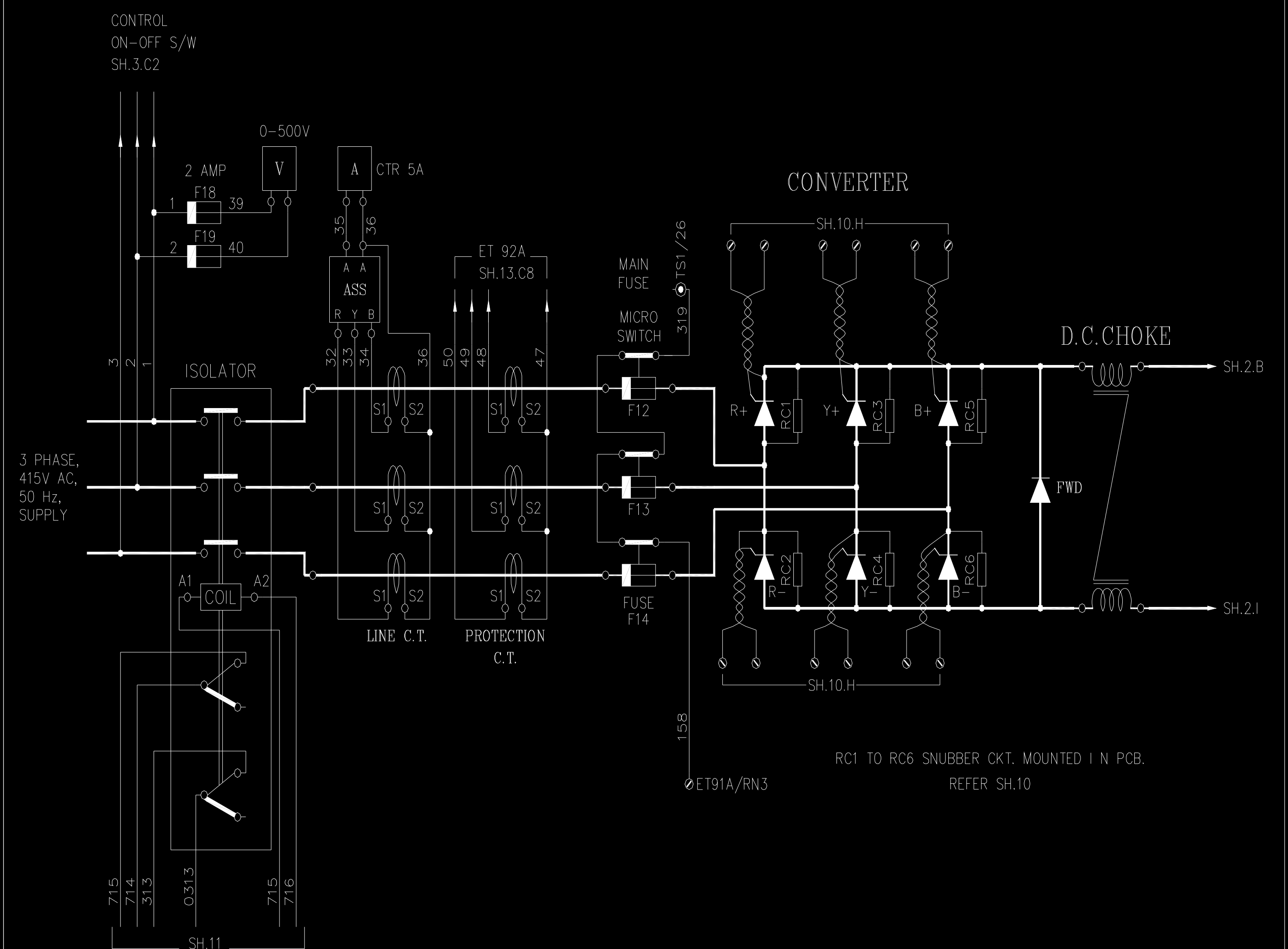
- **ET-Induction Melting Furnace constitute following part:**
  - Furnace Transformer
  - Generator Panel
  - DC Choke
  - Changeover Switch
  - Capacitor Rack
  - De-Mineralise Water Circulation Unit
  - Hydraulic Unit
  - Coil Cradle Assembly
  - Tilting Structure

FURNACE TRANSFORMER

- It is heavy duty Oil Cooled Step Down Transformer meant for supplying power to Induction Furnace.
- Primary side input is Delta connected with 33KV/11KV. Secondary side consist of single or double output of 1000V or 850 V, one as Star output and other as Delta output (only in case of Dual Converter).
- It is of Rectifier Duty Copper Wound type.
- The impedance of furnace allowed is 6.5%, and having 7 offload tap varying +-2.5% each I.e 0 to +7.5% and 0 to –7.5% . Cooling is of type Oil Natural Air Natural.
- All safety gazette are incorporated to trip the Circuit breaker for any fault in transformer. Tripping of Buchhloz Relay, Oil temp. , Coil temperature and Earth Fault are used to ensure protection of furnace transformer in any undesired condition.

GENERATOR PANEL

- **Generator Panel consist of following section:**
  - 3-phase Bridge Rectifier with freewheel diode.
  - (DC to AC) Inverter section
  - Starting Circuit
  - Control Electronic PCB Cards section
  - Transformer for power supply



08

250-600KW_4THY_WO_REMOTE

		NAME	DATE
BISTWER CIRCUIT DIAGRAM	DRN	SP	01/6
	CHD		
	APPD		

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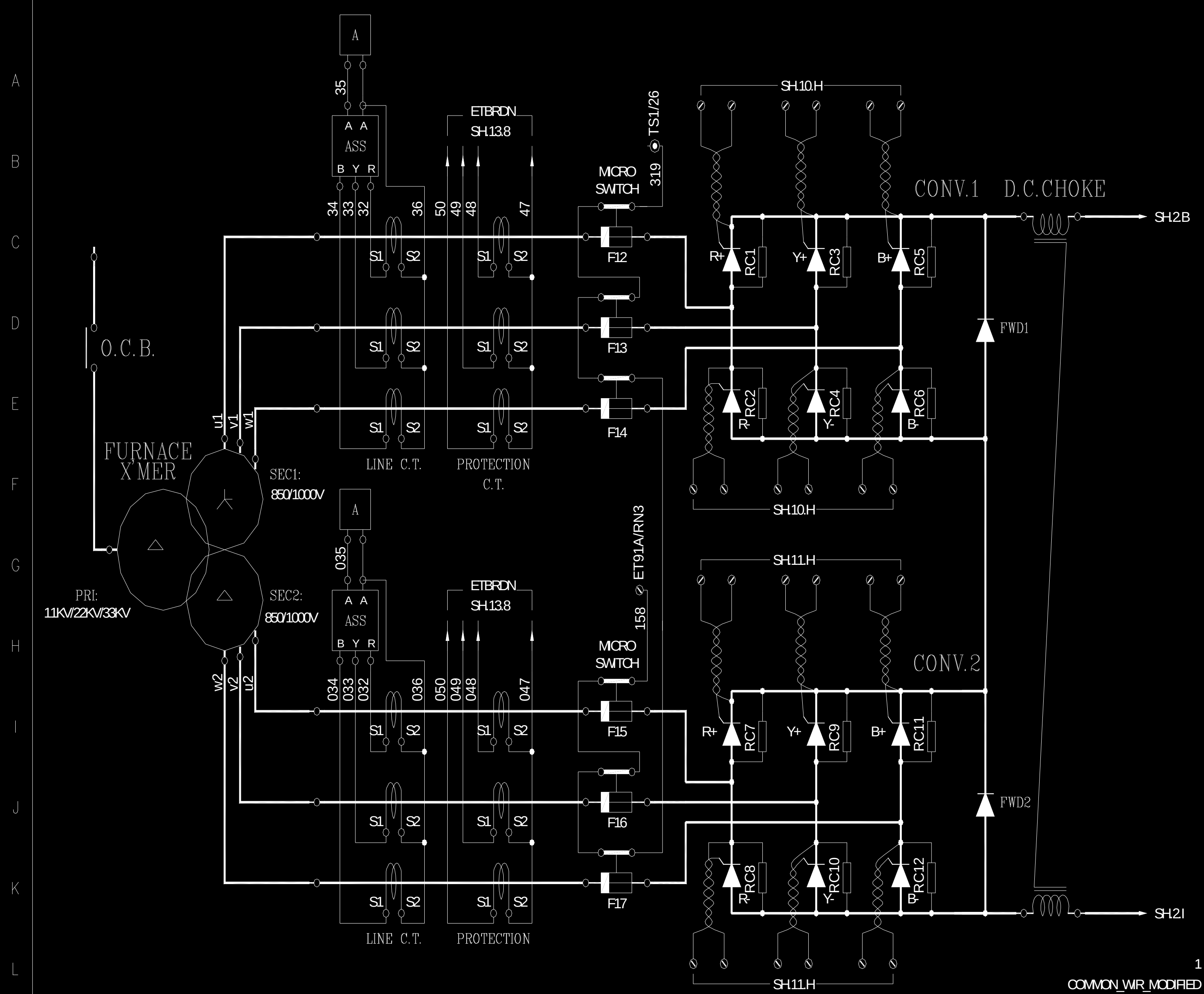
DRG. No. 00 902 00 00 06 31

SHEET 08 OF 18



1

COMMON_WIR_MODIFIED

POWER CIRCUIT DIAGRAM1

	NAME	DATE
DRN	SP	01/6
CHD		
APPD		

ELECTROTHERM INDIA LTD. AHMEDABAD

DRG. No. 0090000000141

SHEET 1 OF 23


QM STAR

3 - f BRIDGE RECTIFIER SECTION

- Incoming mains 3- ϕ supply from furnace transformer is rectified by Full Control Bridge Rectifier.
- Six identical Thyristor which are connected in Bridge configuration to convert Incoming AC to Corresponding DC. Six Thyristor stand for R+, R-, Y+, Y-, B+ & B-.
- Thus the output consist of Six Rectified pulses. One positive and one negative combination of thyristor are fired in accordance to synchronism with incoming main frequency supply.
- Each thyristor are naturally commutated when the supply across it approaches zero. Firing sequence used is (R+ Y-), (R+ B-), (Y+ B-), (Y+ R-), (B+ R-), (B+ Y-).
- Thyristor are used in phase control mode and therefore by varying firing angle, the output power can be varied from lower level to higher level.

3 - f BRIDGE RECTIFIER SECTION ..Cont.

- R C (Resistor and Capacitor) circuit known as snubber circuit is used in parallel to individual thyristor . Snubber circuit protect the thyristor against high voltage spike (dv/dt), which may result in permanent damage to thyristor.
- Full proof and efficient system of control and firing PCB cards are used to give maximum efficiency at all level of operation .
- Transformer T-9F , PCB Card ET-Filter, ET-133/2 and ET-133/3 forms control section for firing of Converter Thyristor.
- Transformer T-9F Step Down incoming 3-Phase from 415 V to 8.66 V.
- ET- Filter PCB cards filters out any high frequency and unwanted noise component from Voltage signal from T-9F. Output is fed to ET-133/2 card.
- ET-133/2 card produce gate pulses in synchronism with incoming mains.

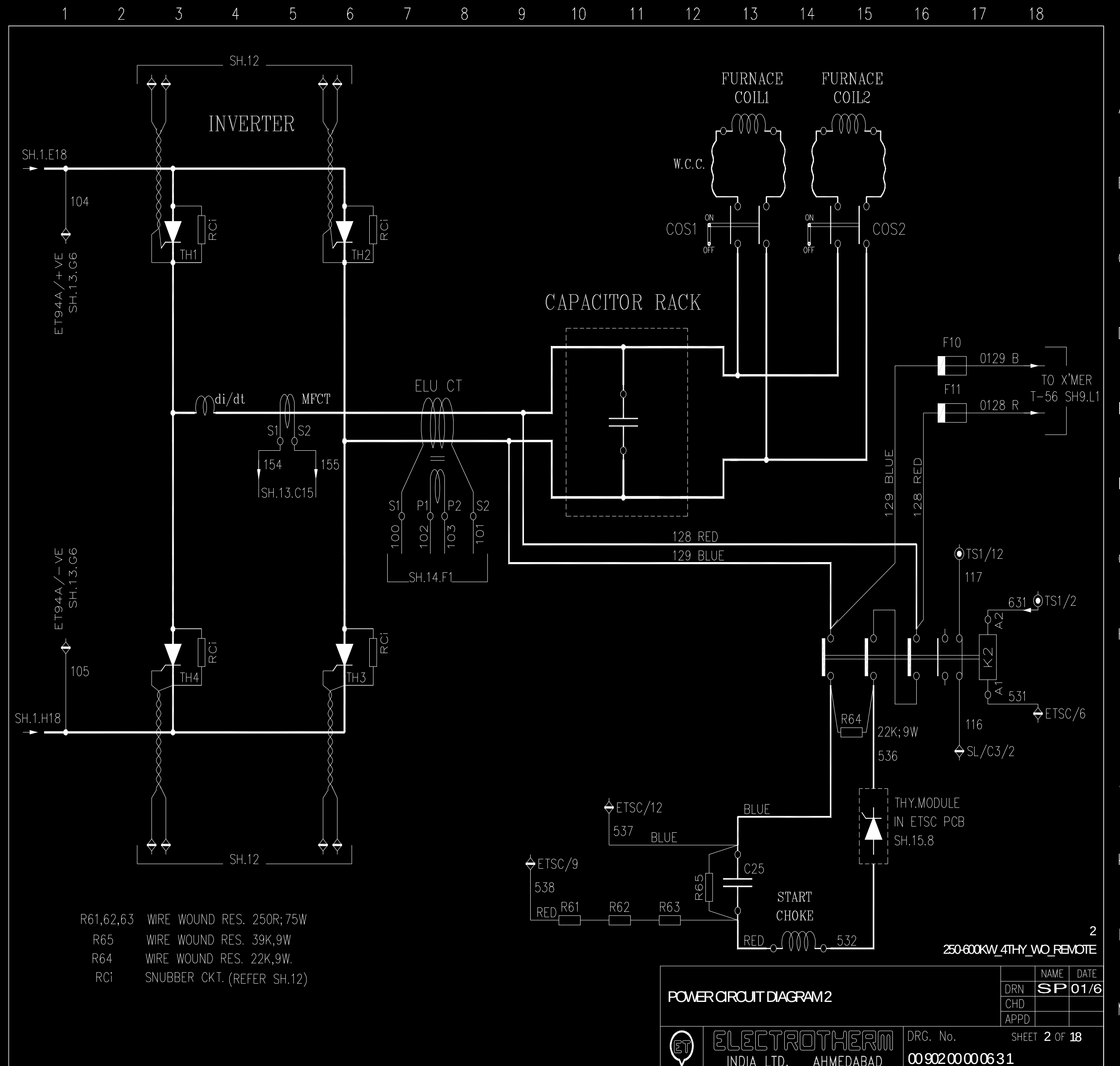
3 - f BRIDGE RECTIFIER SECTION ..Cont.

- Six pulses produce is applied to ET-133/3 card where it is amplified and isolated and fed to thyristor section.
- Individual thyristor are mounted on Water cooled copper base, to absorb the localized heat on the surface of thyristor due to switching losses and thus prolonging the life of costly thyristor.

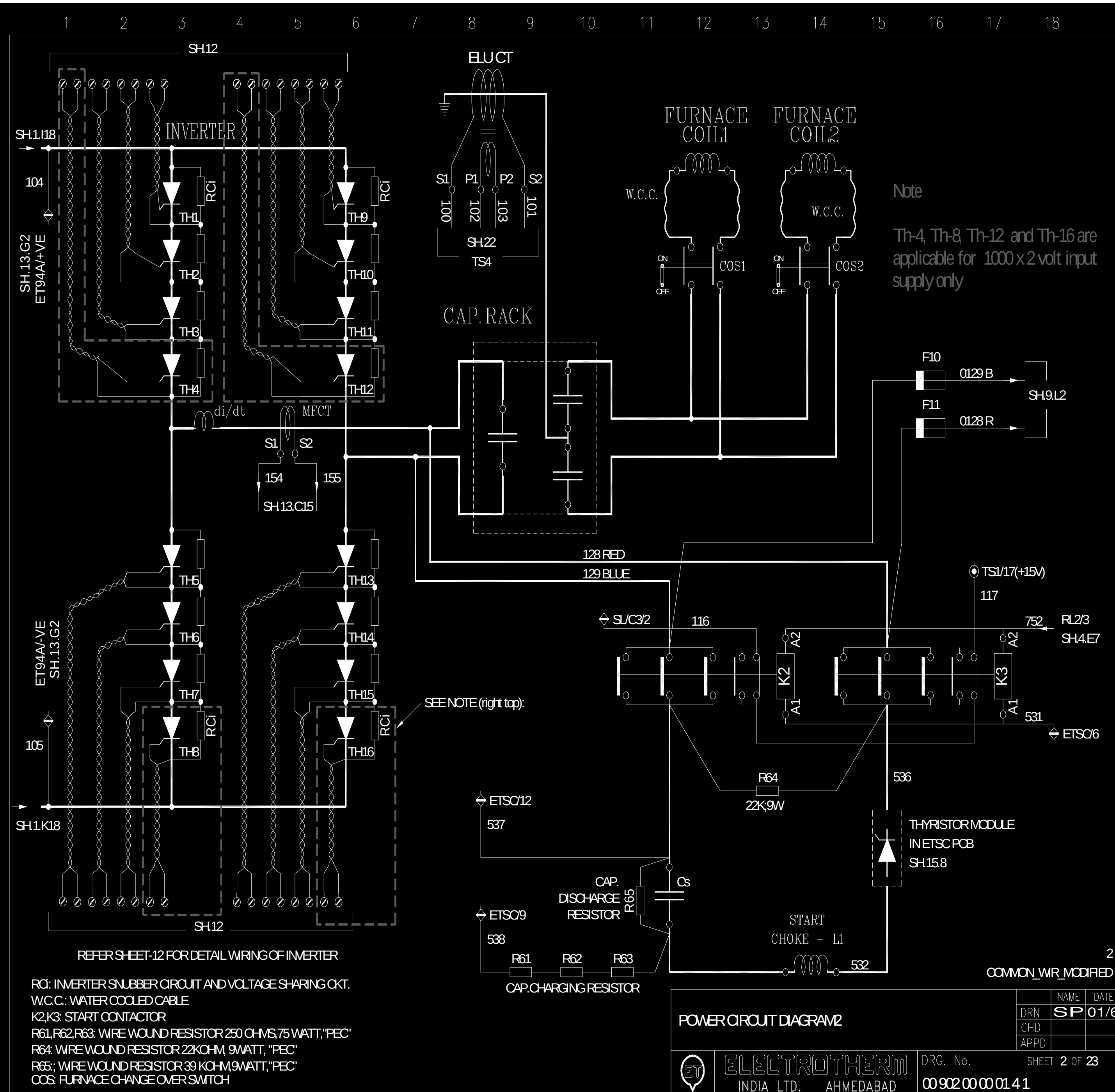
3 - ϕ Bridge Rectifier Section

INVERTER (DC to AC)

- The inverter is responsible for converting High Voltage DC to AC to feed it to the load circuit . It is a single phase bridge connection of four current carrying arms. Configuration used is called Parallel Inverter , as load viz Induction Melting Coil and Capacitors are Parallel to each other.
- Thyristor are connected in four arms. Each arm has serially stack thyristor in range of 3-4 depending of output Voltage designed. Thyristor in diagonal pair are fired in each half cycle of medium frequency , this frequency is determine by parallel resonant LC circuit.
- MF Feedback from Load circuit is tracked by control circuit to generate firing pulses for inverter circuit. ET–ITC PCB card is used for control of Inverter section. Firing signal is send to ET-77 card for amplification and isolation.
- The load circuit is parallel resonant circuit and once energised will have a sinusoidal voltage waveform across it.



INVERTER
SH.12
SH.1.E18
104
ET94A/+VE SH.13.G6
RCi
TH1
TH2
TH3
TH4
ET94A/-VE SH.13.G6
105
SH.1.H18
di/dt
MFCT
S1 S2
154
155
SH.13.C15
ELU CT
S1 P1 P2 S2
100 102 103 101
SH.14.F1
CAPACITOR RACK
FURNACE COIL1
FURNACE COIL2
W.C.C.
COS1
COS2
ON
OFF
F10
F11
0129 B
0128 R
TO X'MER T-56 SH9.L1
129 BLUE
128 RED
128 RED
129 BLUE
TS1/12
117
631
TS1/2
A2
K2
A1
531
ETSC/6
116
SL/C3/2
R64
22K;9W
536
THY.MODULE IN ETSC PCB SH.15.8
ETSC/12
537
BLUE
BLUE
ETSC/9
538
RED
R61
R62
R63
R65
C25
START CHOKE
RED
532
R61,62,63 WIRE WOUND RES. 250R;75W
R65 WIRE WOUND RES. 39K,9W
R64 WIRE WOUND RES. 22K,9W.
RCi SNUBBER CKT. (REFER SH.12)
2
250-600KW_4THY_WO_REMOTE
POWER CIRCUIT DIAGRAM 2
NAME DATE
DRN SP 01/6
CHD
APPD
ELECTROTHERM
INDIA LTD. AHMEDABAD
DRG. No.
SHEET 2 OF 18
00902000006 31



SH12
INVERTER
SH.1.I18
104
SH.13.G2
ET94A/+VE
RCi
TH1
TH2
TH3
TH4
TH9
TH10
TH11
TH12
ELU CT
S1 P1 P2 S2
100 102 103 101
SH22
TS4
FURNACE COIL1
FURNACE COIL2
W.C.C.
COS1
COS2
ON
OFF
Note
Th-4, Th-8, Th-12 and Th-16 are applicable for 1000 x 2 volt input supply only
CAP. RACK
di/dt
MFCT
S1 S2
154 155
SH.13.C15
F10
0129 B
F11
0128 R
SH.9.L2
128 RED
129 BLUE
TS1/17(+15V)
117
SL/C3/2
116
752
RL2/3
SH.4.E7
A2
K2
A1
K3
531
ETSC/6
TH5
TH6
TH7
TH8
TH13
TH14
TH15
TH16
ET94A/-VE
SH.13.G2
105
SEE NOTE (right top):
SH.1.K18
R64
22K,9W
536
THYRISTOR MODULE
IN ETSC PCB
SH.15.8
ETSC/12
537
CAP.
DISCHARGE
RESISTOR
R65
Cs
ETSC/9
538
START
CHOKE – L1
532
R61 R62 R63
CAP.CHARGING RESISTOR
REFER SHEET-12 FOR DETAIL WIRING OF INVERTER
RCi: INVERTER SNUBBER CIRCUIT AND VOLTAGE SHARING CKT.
W.C.C.: WATER COOLED CABLE
K2,K3: START CONTACTOR
R61,R62,R63: WIRE WOUND RESISTOR 250 OHMS,75 WATT,"PEC"
R64: WIRE WOUND RESISTOR 22KOHM, 9WATT, "PEC"
R65.: WIRE WOUND RESISTOR 39 KOHM,9WATT,"PEC"
COS: FURNACE CHANGE OVER SWITCH
2
COMMON_WIR_MODIFIED
POWER CIRCUIT DIAGRAM2
NAME DATE
DRN SP 01/6
CHD
APPD
ELECTROTHERM
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DRG. No.
SHEET 2 OF 23
00 902 00 00 01 4 1

INVERTER (DC to AC) …cont

- Similarly the incoming thyristor cannot be allowed to carry the full current instantaneously. A time has to be given to the current to rise slowly. During this portion all the thyristor are conducting and this is called Over Lap time.
- In the Parallel Resonance circuit, less current is drawn through inverter. Its major current requirement is fulfill by MF capacitor connected in parallel to Inductor(Coil). So lower capacity of thyristor are required for higher KW furnace.
- As Stacks of serially connected Thyristor are used in every arm of inverter , uniform voltage sharing is required across each thyristor. This is fulfilled by fixing Resistor across each thyristor . Also RC Snubber circuit is used to to suppress any voltage spike produce during switching.

INVERTER (DC to AC) …cont

- As the current is flowing from a constant current supply, the output current has got a square waveform.
- Forced commutation of the conducting thyristor is ensured by firing the incoming thyristors at a point in advance of the voltage zero crossing, such that a instantaneous voltage across the conducting thyristors is reversed to cause their commutation.
- As the current is flowing from a constant current supply, the output current has got a square waveform.
- A thyristor will not sustain forward voltage after commutation till a short period elapses. This period is called turn off time(TOT) and must be made available to the thyristor before the approaching zero crossing.

Inverter Section

STARTING CIRCUIT

- Starting circuit plays major role for reliable starting of furnace.
- It consist of Starting Card, Start Capacitor, Start Choke, Start Thyristor Module and Start Contactor unit.
- Voltage output from T-17 is Rectified to DC voltage and send to start capacitor via charging resistor.
- The function of this card is to commutate the priming pair of Inverter , so that with firing of other diagonal pair , one cycle is completed. This will lead to voltage being build up and furnace to switch ON. Inverter pair will then get commuted by resonance circuit. Thus this circuit is used only once and will be out of focus till next time when furnace will be made on.
- With the pressing of Heat On button , signal is send to this card to Switch On the start contactor. After few millisecond, signal to make contactor Off is send and at same time gate pulses to start thyristor are released.

STARTING CIRCUIT ...cont

- With the switch On of Start Thyristor connection to load circuit is established and bunch of High energy pulses generated due to series combination of Start Choke and Charged Capacitor is injected into the load.This energy oscillate at twice the frequency of load circuit.
- The energy released makes the resonance Circuit Ringing.The magnitude of these energy pulses is large enough to commute the first pair of thyristor .

Starting Section

ELECTRONIC CONTROL CARD

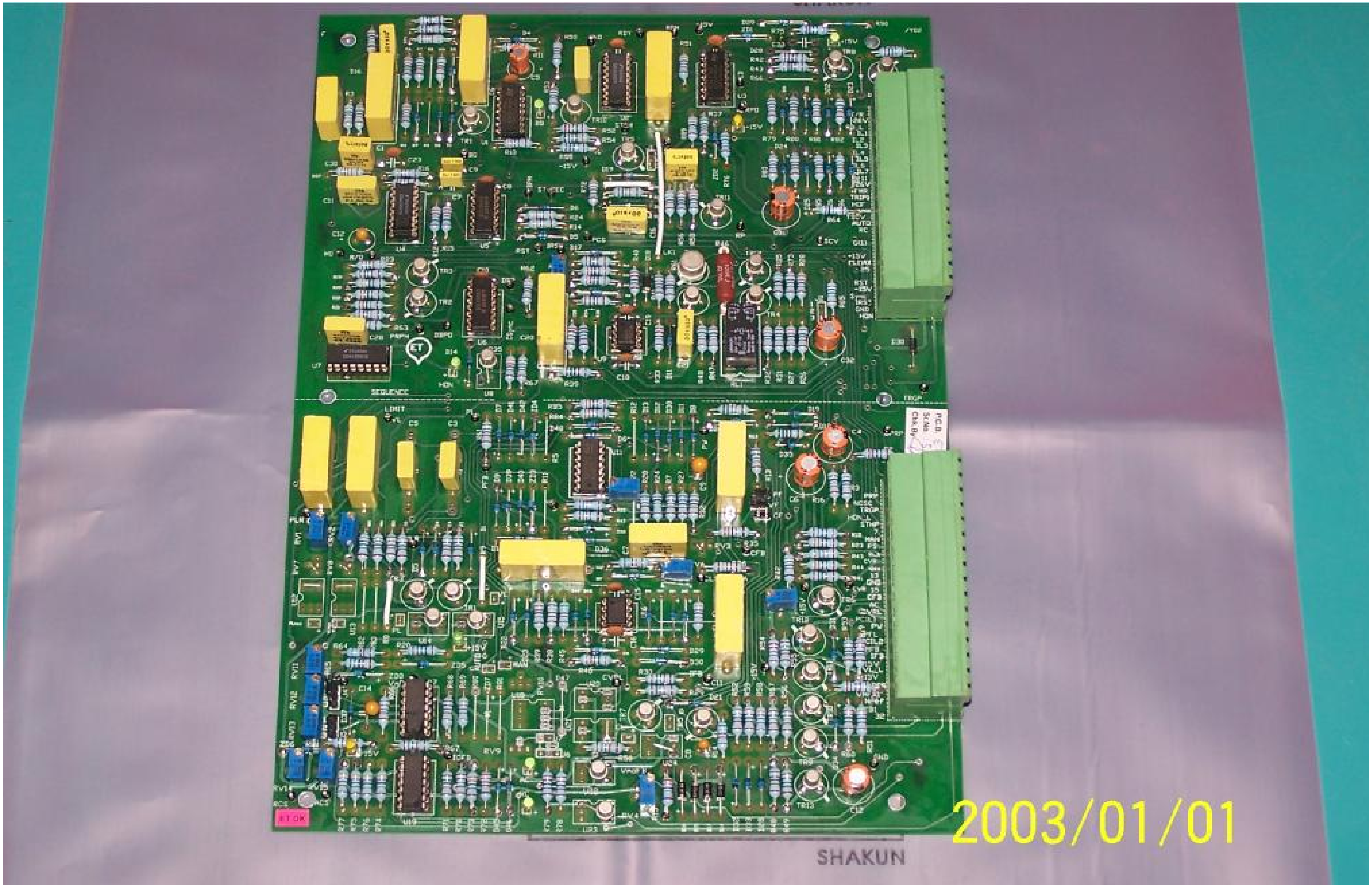
- **This section consist of following PCB cards:**
  - ET-SL (Sequence and Limit Control Card)
  - ET-ITC (Inverter Control Card)
  - ET-TOV (TOT Generation and Over voltage Detection)
  - ET-PS (Power Supply)
  - ET-SC (Starting Card)
  - ET-133/2 (Converter Firing Pulse Generation Card)
  - ET-133/3 (Amplification and Isolation Card)
  - ET-77 (Inverter Firing Amplification and Isolation)
  - ET-84 (Water Conductivity Measure)
  - ET-125 (Earth Leakage Detection)
  - ET-BRDN (Burden Board)
  - ET-RLY (Relay)

ET-SL
(SEQUENCE AND LIMIT)

- This is one of the important card, responsible for start and stop of generator in proper sequence. For this card to work Ready Signal is required to be healthy.
- Ready signal is high only when all interlocks viz Door, DM Water flow, DM water temperature, Fuse blown , Water conductivity, Earth leakage and other External Parameter like Coil water temperature, Coil water pressure and Outgoing Busbars Temperature are in healthy condition.
- This Ready signal is required to make all condition conducive for safe working of furnace. If any of the interlock fails Furnace gets trip and visual display is lit of respective interlock for operator to see and rectify.
- With Ready signal on Display , when Heat On button is pressed a reset pulse is generated to bring Digital IC's to there normal condition.

ET-SL ...Cont.

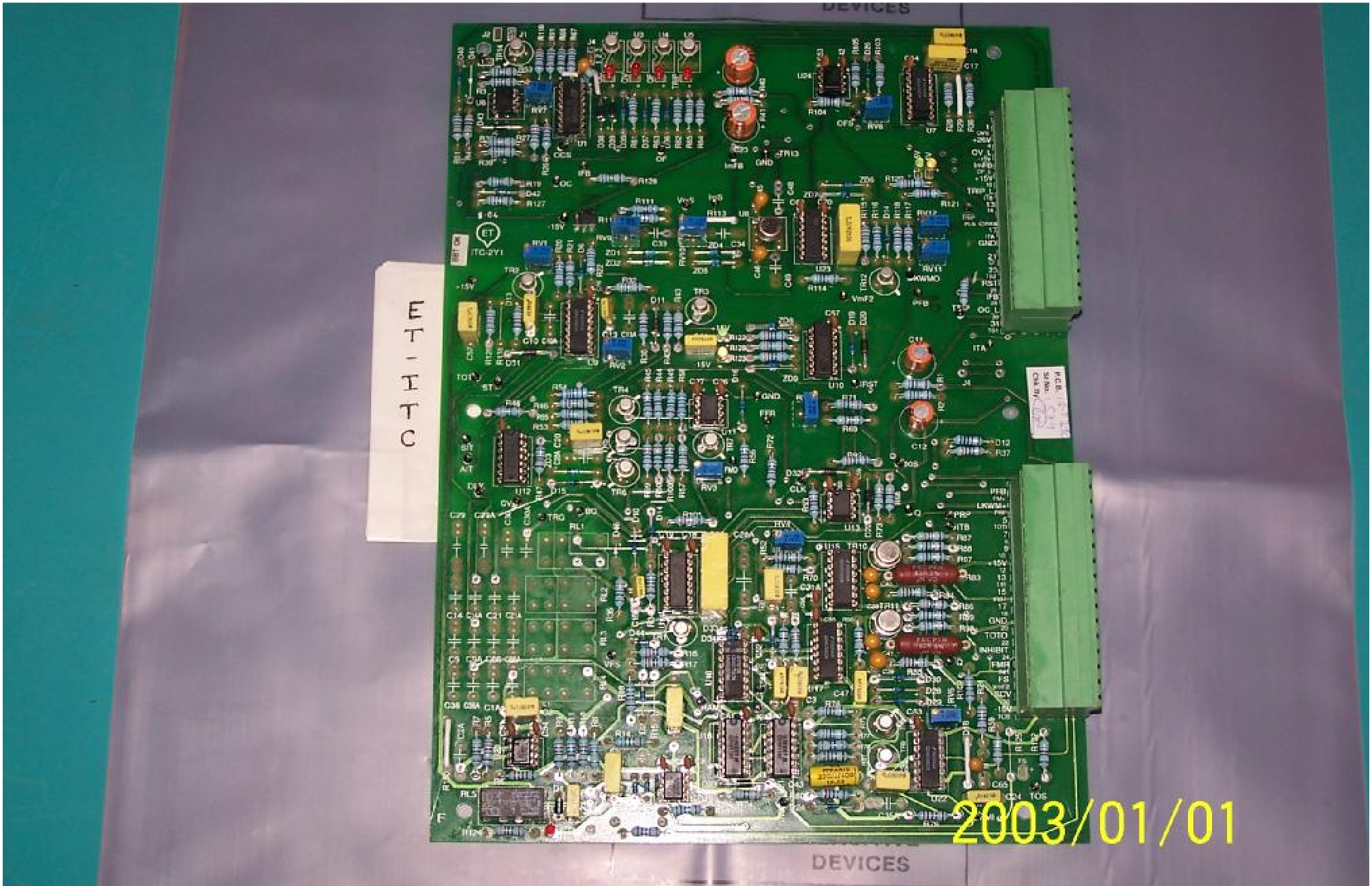
- Control reference voltage is send to converter firing card for converter thyristor to switch on, there after various signal are generated in sequence and some after comparison with feedback to work toward reliable switch ON of furnace
- Power pot variation is sense by this cards to raise or lower the output power as per operator. Control reference voltage is increase if power has to be increased and it is lowered to reduce the power.
- Voltage and Current feedback is taken to compare it with set value so that furnace do not exceed the prescribed limit of power.



ET-SL

ET-ITC
(INVERTOR CONTROL CARD)

- This card is responsible for firing the Inverter thyristor as per the feedback from MF circuit. Frequency changes with load of charge in Coil and also with Melting condition. This feed back is taken from Feedback Transformer which step-down high MF voltage to 100V.
- This MF voltage is utilize in card to produce firing pulses corresponding to its peak , so that inverter is fired for its full capacity. Also this MF signal is used along with MF current for display and calculation of Actual KW and frequency.
- Actual TOT signal from ET-TOV card is sensed and is compared with Set TOT . The difference if any makes corresponding shift in firing of Inverter thyristor to take into account overlap and TOT requirement.
- Over Current, Over Voltage and Over Frequency trip are generated in this circuit.



ET-ITC

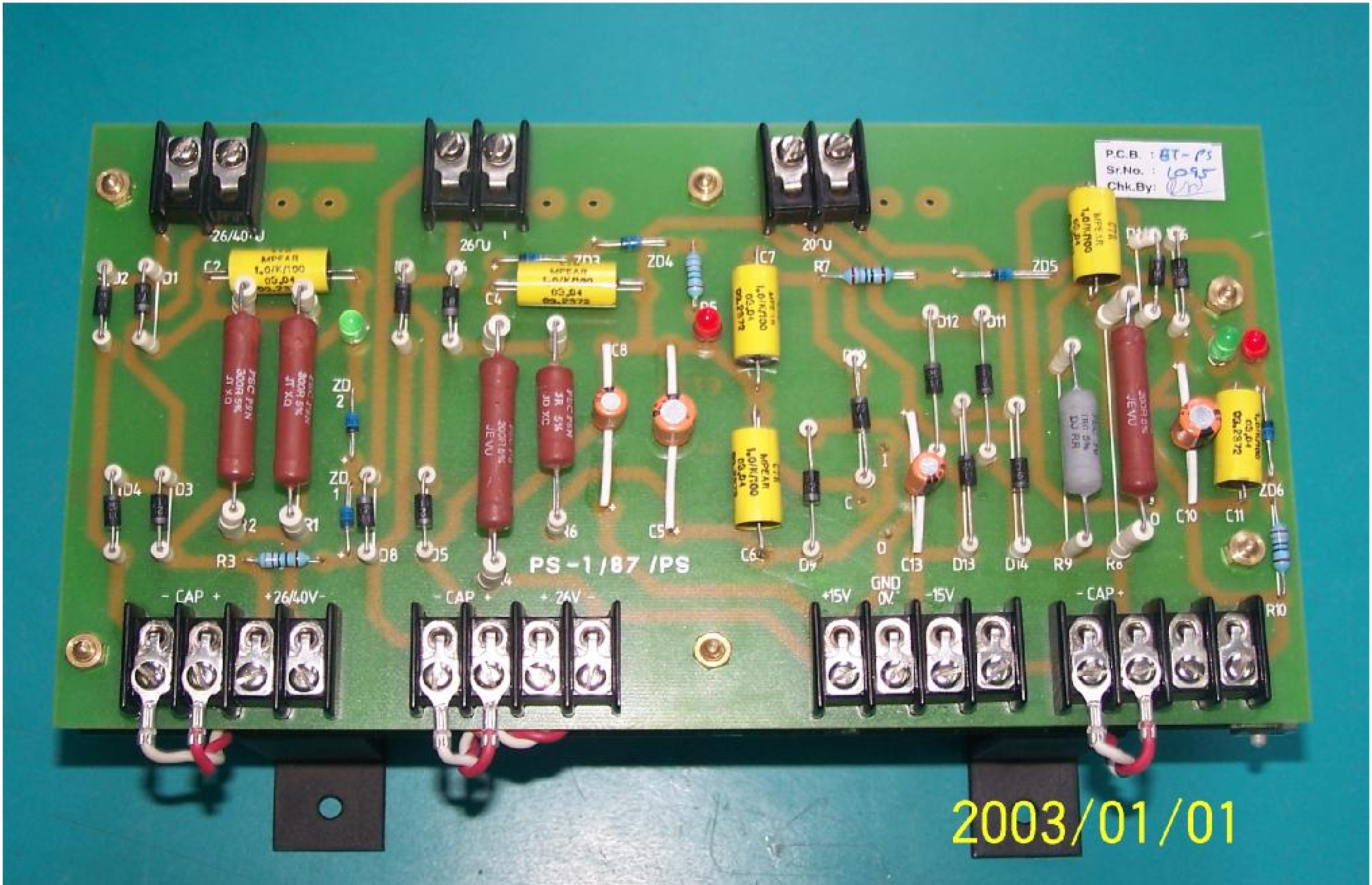
ET-TOV
(TOT & OVER VOLTAGE)

- Input to this card is High Voltage DC from input of Inverter.
- This high voltage is potentially divided to lower voltage. The signal thus received is used to generated Pulses during the time period when inverter thyristor is reverse biased. This is TOT signal.
- TOT signal generated is fed to ET-ITC card, where it is compared with set TOT to fire the Inverter so that TOT remain with in prescribed limit.
- Over Voltage is detected and corresponding signal is generated if incoming DC Voltage is higher than set Limit. This OV signal is fed to ITC for tripping of generator.

ET-TOV Card (94)

ET-PS (POWER SUPPLY)

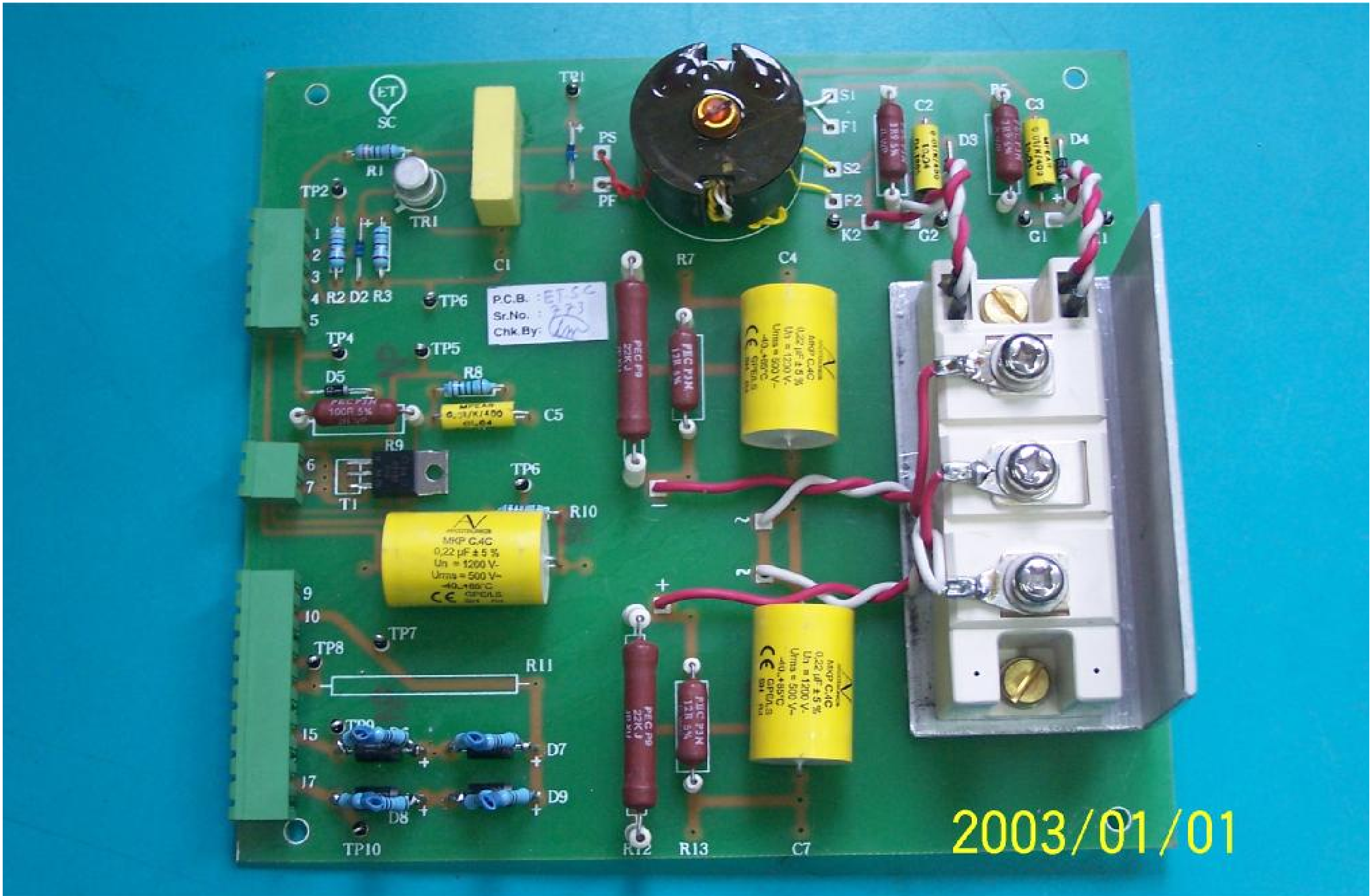
- Function of this card is to Rectify the input AC to generate Regulated and Unregulated DC Voltage to be used in various PCB card and control circuit in generator panel.
- Input AC voltage is fed from Constant Voltage Transformer (T-12) . Output Regulated Dc +15V, Gnd & -15V is supplied to all Logic Cards.
- Unregulated DC voltage +26V & Gnd is supplied for Relays, Interlocks, Display Lamp and for Amplification of Firing Gate Pulses.


P.C.B. : ET-PS
Sr.No. : 1095
Chk.By:
PS-1/87/PS
- CAP +
+26/40V-
+ 26V -
+15V
GND 0V
-15V
2003/01/01

ET- PS

ET-SC (STARTING CARD)

- This cards operate under command of ET-SL card.
- ET-SC cards is responsible for ON & OFF of start Contactor , Charging of Start Capacitor & firing of Start Thyristor.
- AC voltage fed from T-17 transformer is rectified by this cards and send to start capacitor through charging resistor.
- ET-SL give the signal to operate the start contactor by firing Triac in ET-SC card. ET-SL also send signal to generate gate pulses for firing of start thyristor associated with ET-SC card.
- Function of this card is to inject high amplitude energy pulses in resonance circuit (Load - LC, so that it start ringing and helps in commutation of first pair of Inverter Thyristor.
- ET-SC comes in role only once that too at the beginning of Heat ON cycle. Once the panel is delivery output power this card is isolated from main circuitry.


ET
SC
TR1
PS
PF
S1
F1
S2
F2
K2
G2
G1
C2
D3
R5
C3
D4
R1
TP2
TR1
C1
R2 D2 R3
TP6
P.C.B. : ET-SC
Sr.No. :
Chk.By:
R7
C4
TP4
TP5
D5
R8
C5
R9
T1
TP6
R10
MKP C4C
0,22 µF ± 5 %
Un = 1200 V-
Urms = 500 V~
TP7
TP8
R11
D7
D9
D8
TP10
R12
R13
C7
2003/01/01

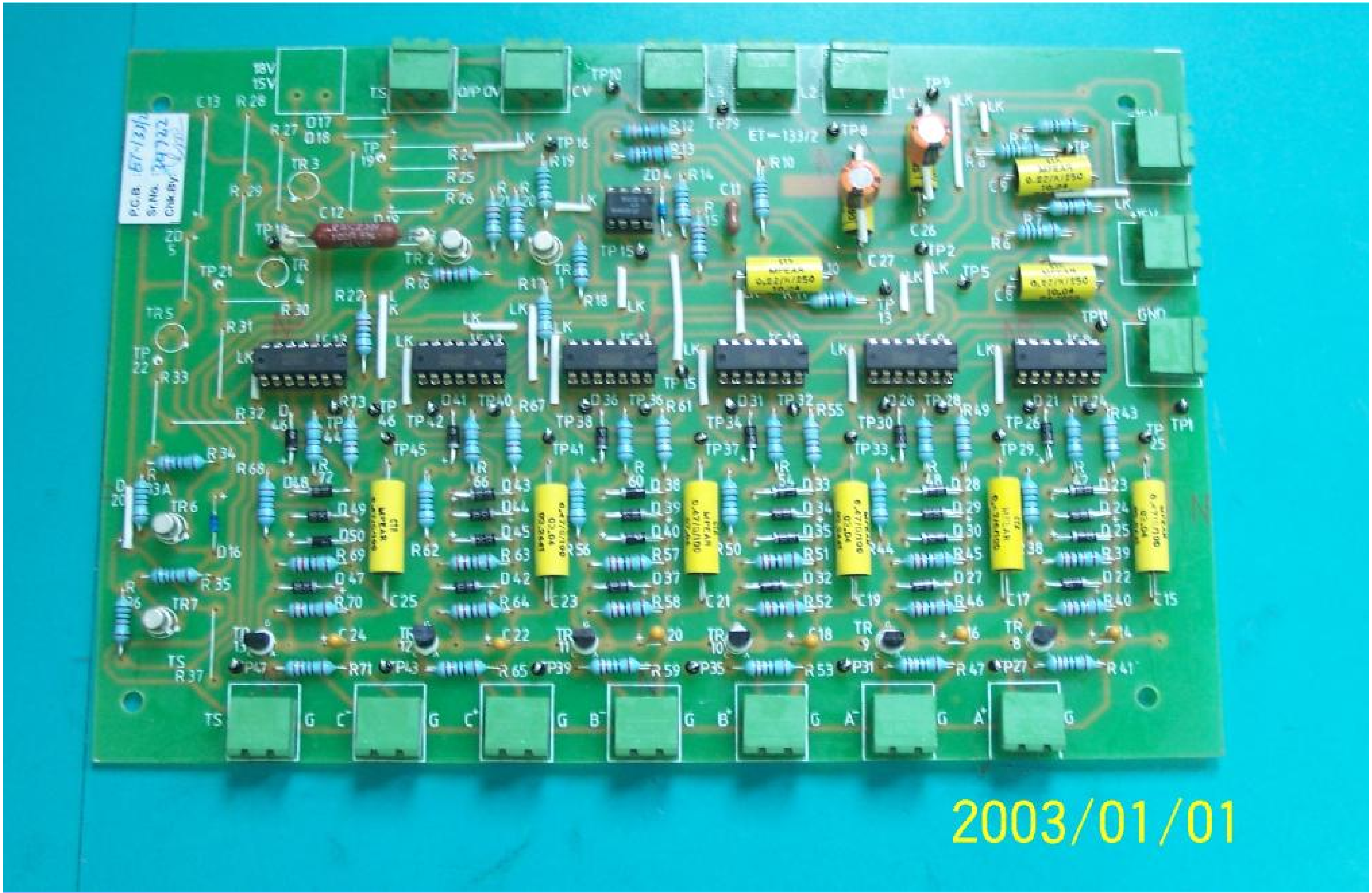
ET-SC

ET-133/2 (CONVERTER FIRING PULSE GENERATION)

- The function of ET-133/2 card is to generate firing gate pulses for Thyristor in 3-phase Fully Controlled Bridge Rectifier.
- Firing angle for converter thyristor has to be shifted in accordance to Power Pot to deliver Minimum to Maximum output power.
- Six identical analog IC blocks and logical electronic circuit are used to generate gate pulses for six converter thyristor. IC blocks detects Zero crossing of 3-phase signal and convert positive and negative cycle of each phase to square wave. These Square wave is then converted into Ramp wave.
- Ramp wave thus generated is compared with Control Reference DC Voltage from ET-SL card to generate a positive pulse at point of interaction.

ET-133/2 …cont

- This Reference DC voltage Signal is varied in accordance to Power Pot, and hence the point of intersection varies and so is the firing angle.
- Trip circuit is incorporated in ET-133/2 card which on receive of trip signal from ET-ITC card ground all the six gate pulses.



ET-133/2

ET-133/3
(AMPLIFIER & ISOLATION)

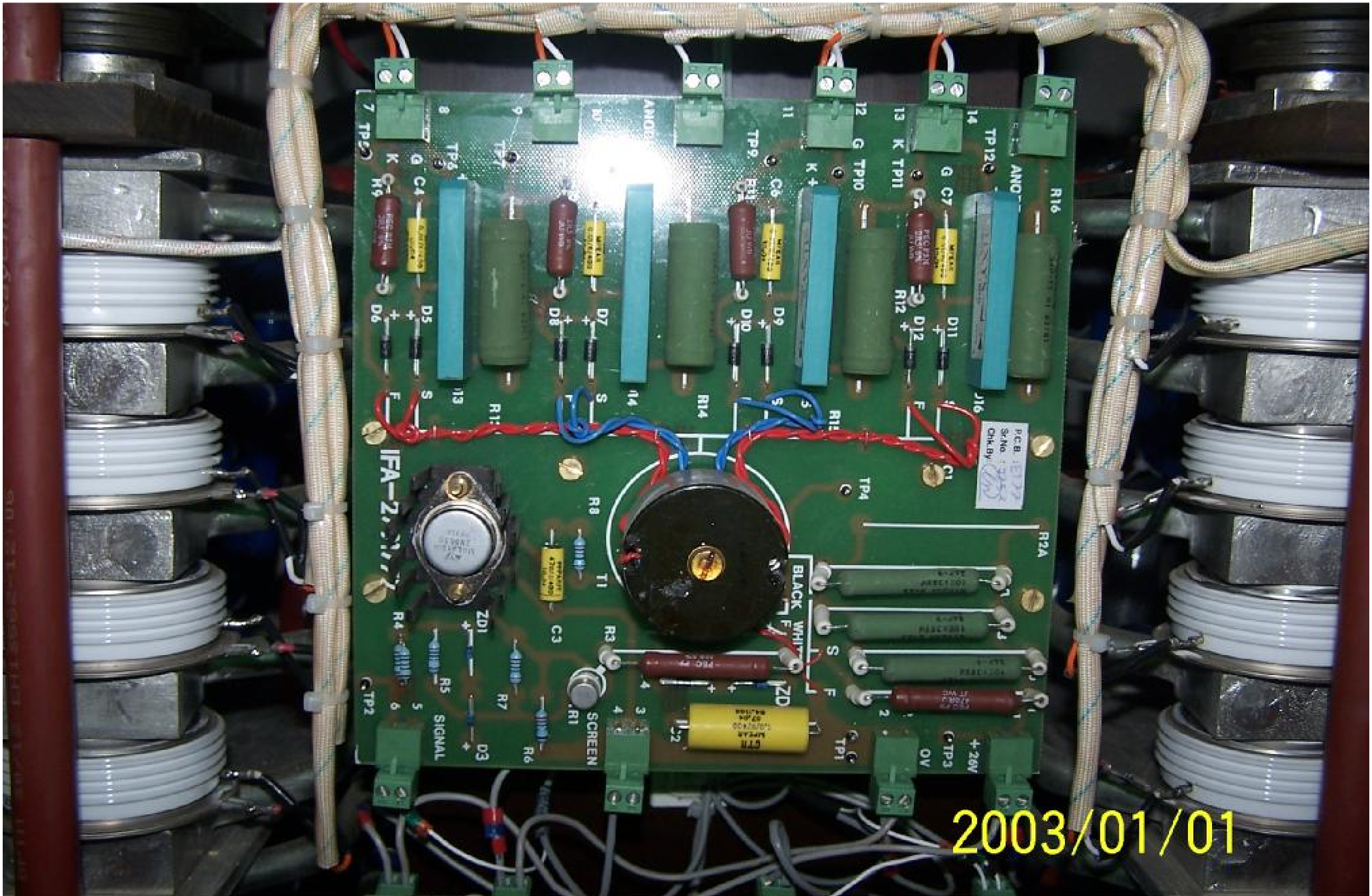
- Six Gate pulses and Trip signal from ET-133/2 card are fed to ET-133/3 card .
- The function of ET-133/3 is , amplification of gate pulse required for reliable switching of Thyristor to ON condition.
- After suitable amplification it is isolated using Pulse transformer. Isolation is utmost required ,so that any disturbance from thyristor end is not reflected to control card.
- Necessary circuit are incorporated for safe guarding against pick of noise signal and thus any misfiring of thyristor at inappropriate time is ruled out.
- The output gate pulses are connected to individual Thyristor Gate with respect to Cathode.

ET-133/3

ET-77 (INVERTER FIRING AMPLIFIER)

- This card receive pulses from ET-ITC card for firing Priming and Non Priming Inverter pair alternately.
- Only diagonally opposite thyristors arm are fired at a particular instance. The current flow in positive direction in One half and in Negative in other half of MF cycle.
- Necessary amplification ,wave shaping and isolation is done in this card.
- Pulse transformer is used for isolation and producing 3 to 4 simultaneous gate pulses at secondary as per number of serially connected thyristor in individual arm.
- The width of pulse used is 40 micro sec ,its peak voltage(Vp) is 5 to 6V and steady state voltage(Vs) is 3 to 4 V.

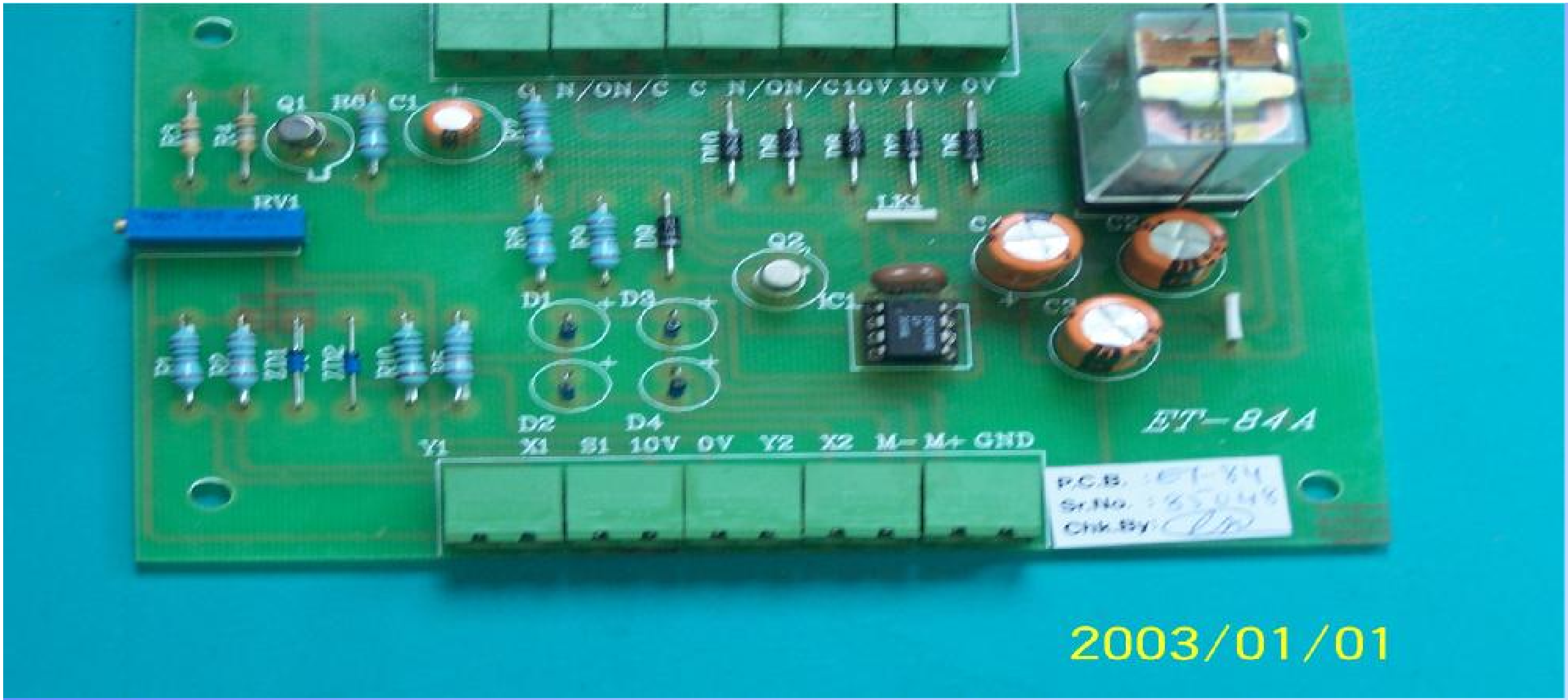
- It also consist of Beak Over Diode(BOD) , which is used primarily for protection of Thyristor. It is connected across thyristor in ET-77 card.Voltage rating for break down of BOD is lower than thyristor , so in case of any fault the diode conduct and thus bypass thyristor and protect it.



ET-77 card

ET-84 (WATER CONDUCTIVITY CARD)

- The power components mounted in the MF Generator dissipates lots of heat. Since thermal charateristic affect the efficient working of component and can lead to malfunction , proper cooling has to be provided.
- Demineralise water having lower conductivity is used to cool the power components. Higher conductivity leads to scaling in copper tubes and base and hence will lower the cooling rate.
- Function of this card is to continuously monitored the conductivity level of water and trip the generator if the conductivity is higher than set value.
- It works on principle of measuring the resistivity across a column of water. Resisitivity decreases with increase in conductivity & vice versa. Sensor across water column form one arm of bridge circuit , change in resistivity changes the balance in circuit and hence the current flow. These change in current flow is detected and display on Conductivity meter and also used in tripping with respect to set value.



G N/ON/C C N/ON/C10V 10V 0V
Q1
C1
RV1
LK1
Q2
IC1
D1
D3
D2
D4
C2
C3
ET-84A
Y1
X1 S1 10V 0V Y2 X2 M- M+ GND
P.C.B.
Sr.No.
Chk.By
2003/01/01

ET-84

Sensor

2003/01/01

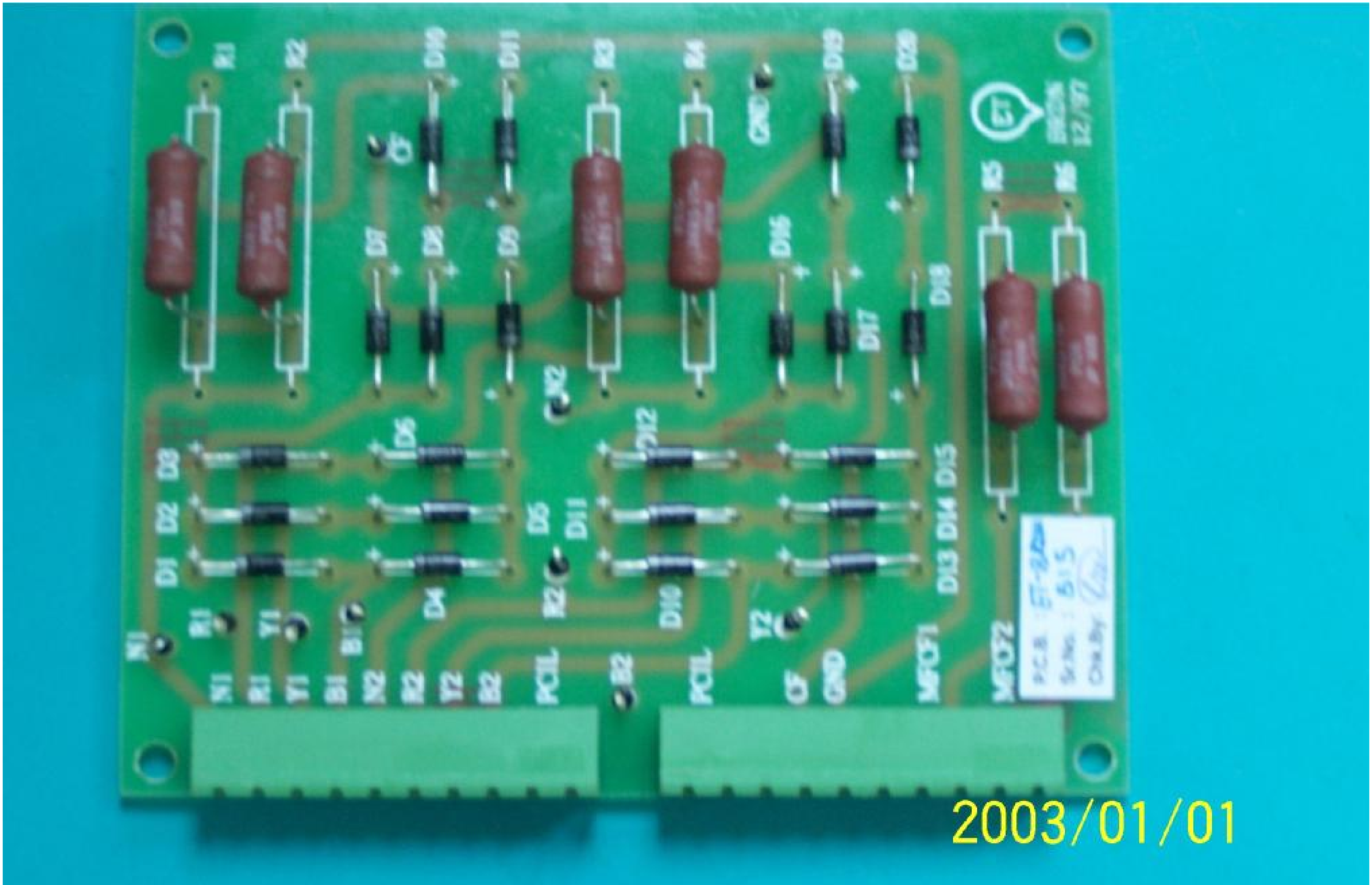
ET-125 (EARTH LEAKAGE DETECTOR)

- The system, which includes both a earth leakage detector circuit associated with the power supply and a earth leak detector probe, located in the furnace, is designed to provide important protection against electrical shock and warning of metal-to-coil penetration, a highly dangerous condition that could lead to a furnace eruption or explosion.
- The principle of operation of this safety circuit is that it detects the difference between the flow of current in incoming and outgoing inverter busbars.If the difference is greater than safe limit the safety circuit trips the power. It continuously measure the the difference as leakage current indication on Ammeter.
- The detection is done by ELU CT connected to MF Busbars. A test facility is also incorporated in this circuit to test the working of this safety circuit.

ET-125 (ELU)

ET-Brdn(BURDEN BOARD)

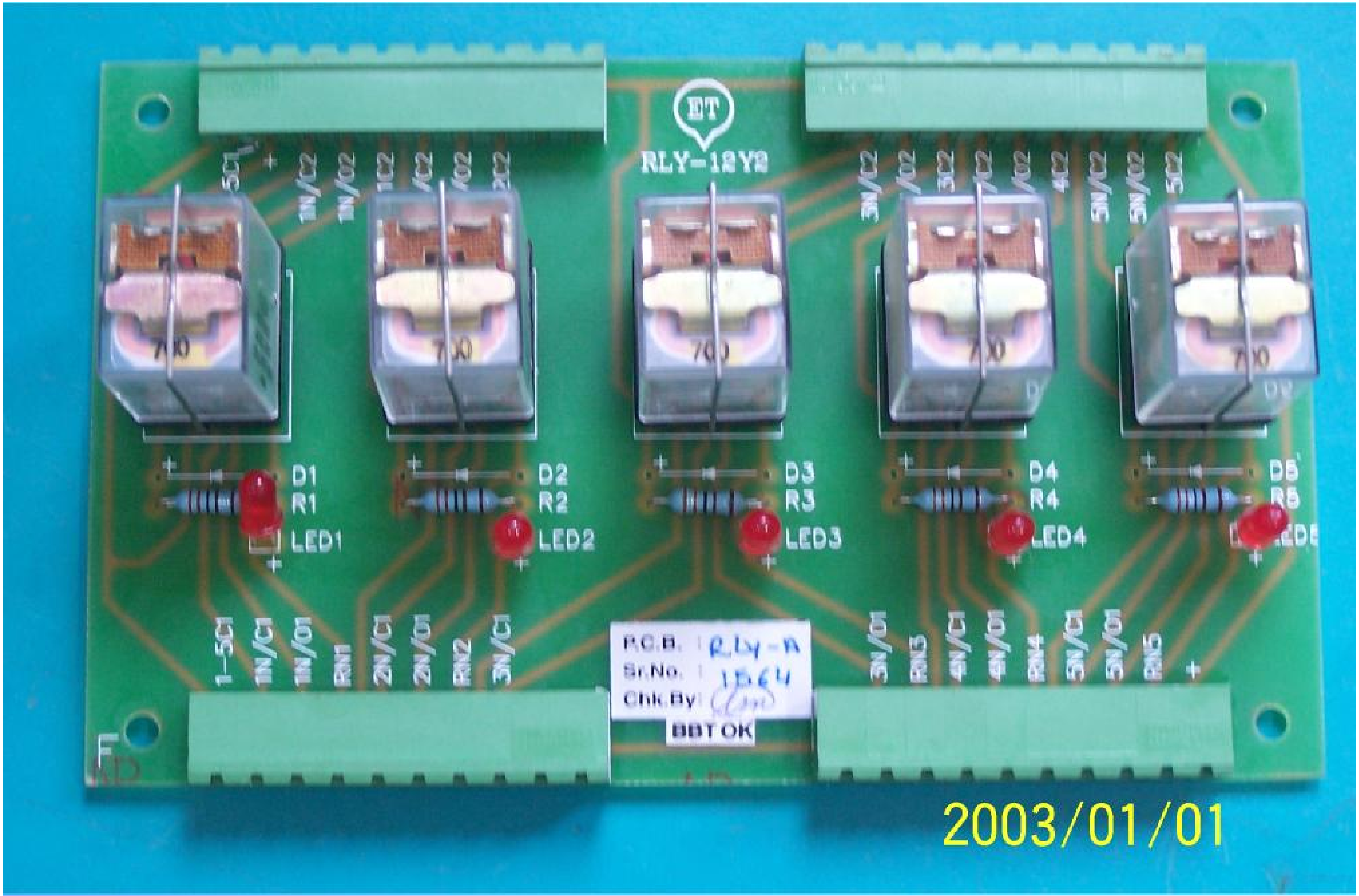
- For generator to operate with in safe limit, current drawn by it has to be measured and compared with set value.
- Incoming Current is measured by Protection CT connected to incoming lines. MF current is measure by MF CT connected to inverter output.
- These AC current is to converted to proportional DC signal for the control circuit to operate.
- ET-Brdn card perform the function of converting AC current signal to DC signal to be send to control circuit for power calculation and protection against short circuit current in case of failure of power components.



ET-Brdn

ET-RLY (RELAY CARD)

- Function of these card is to operate LED Indication related to signal sensed and generated by control cards.
- Some are used for interlocks and some for certain logic.
- Each relay block have 2 sets of NO-NC block.

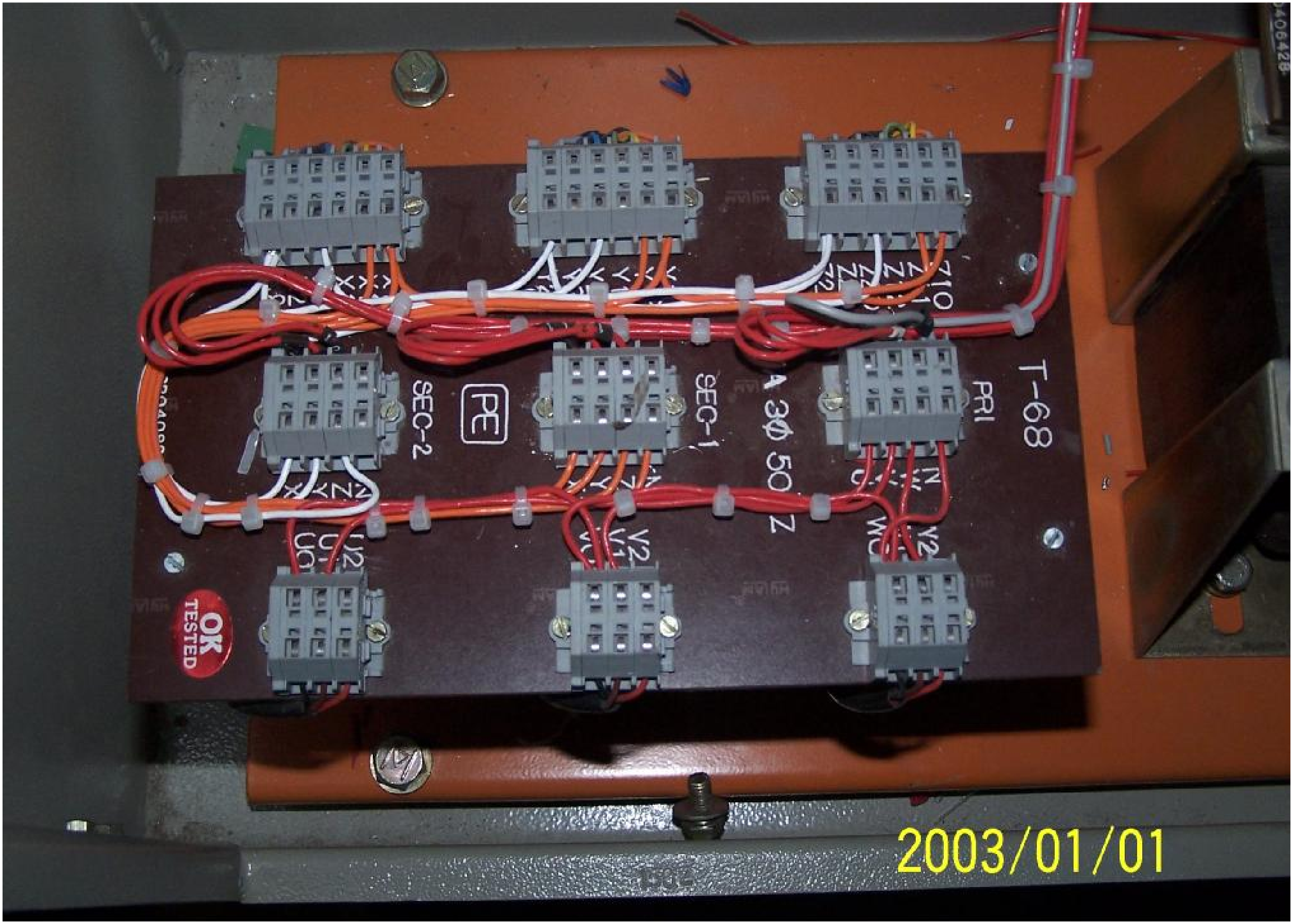

RLY-12Y2
D1
R1
LED1
D2
R2
LED2
D3
R3
LED3
D4
R4
LED4
P.C.B.
Sr.No.
Chk.By
BBT OK
2003/01/01

ET-RLY

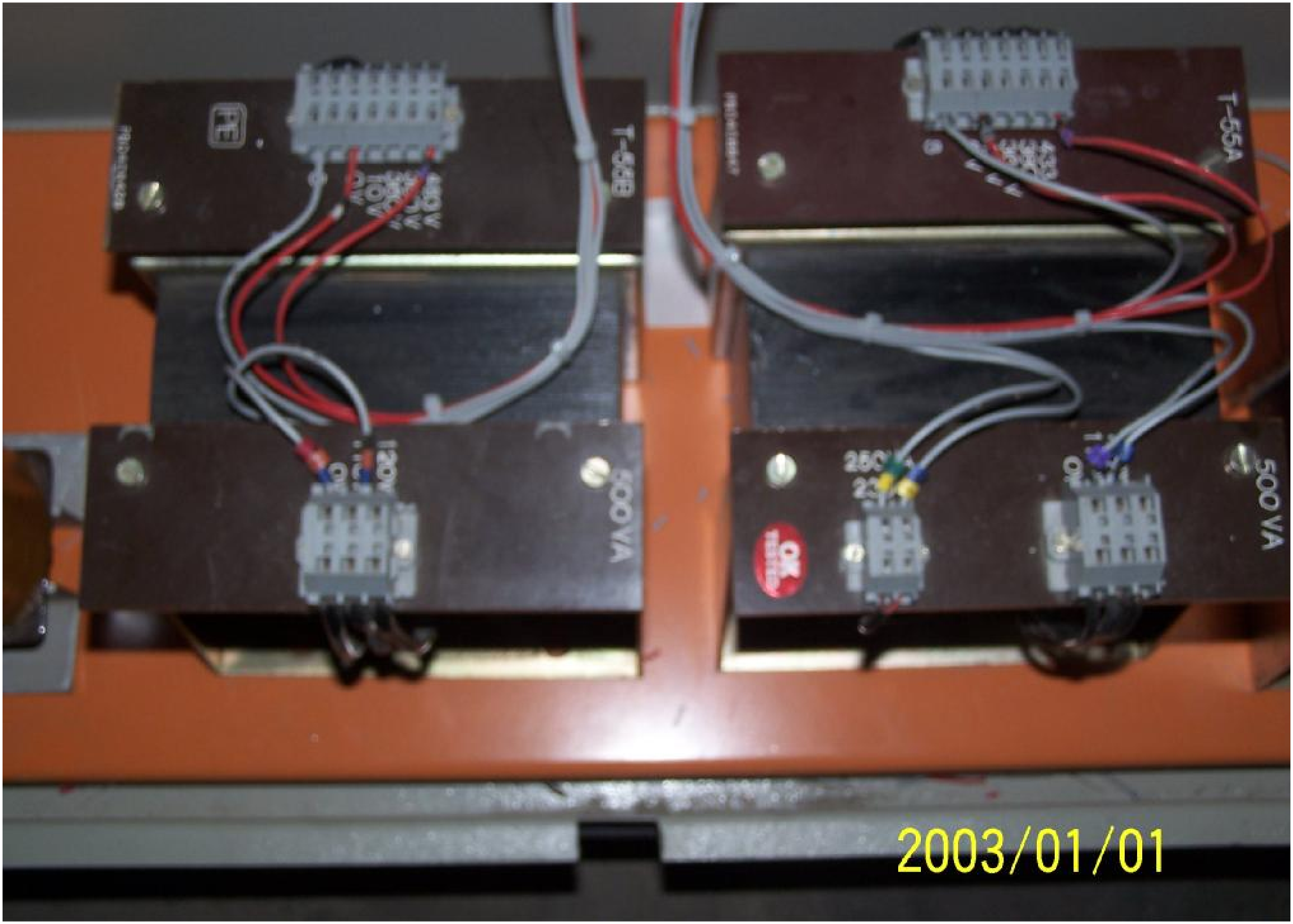
TRANSFORMER FOR CONTROL SUPPLY

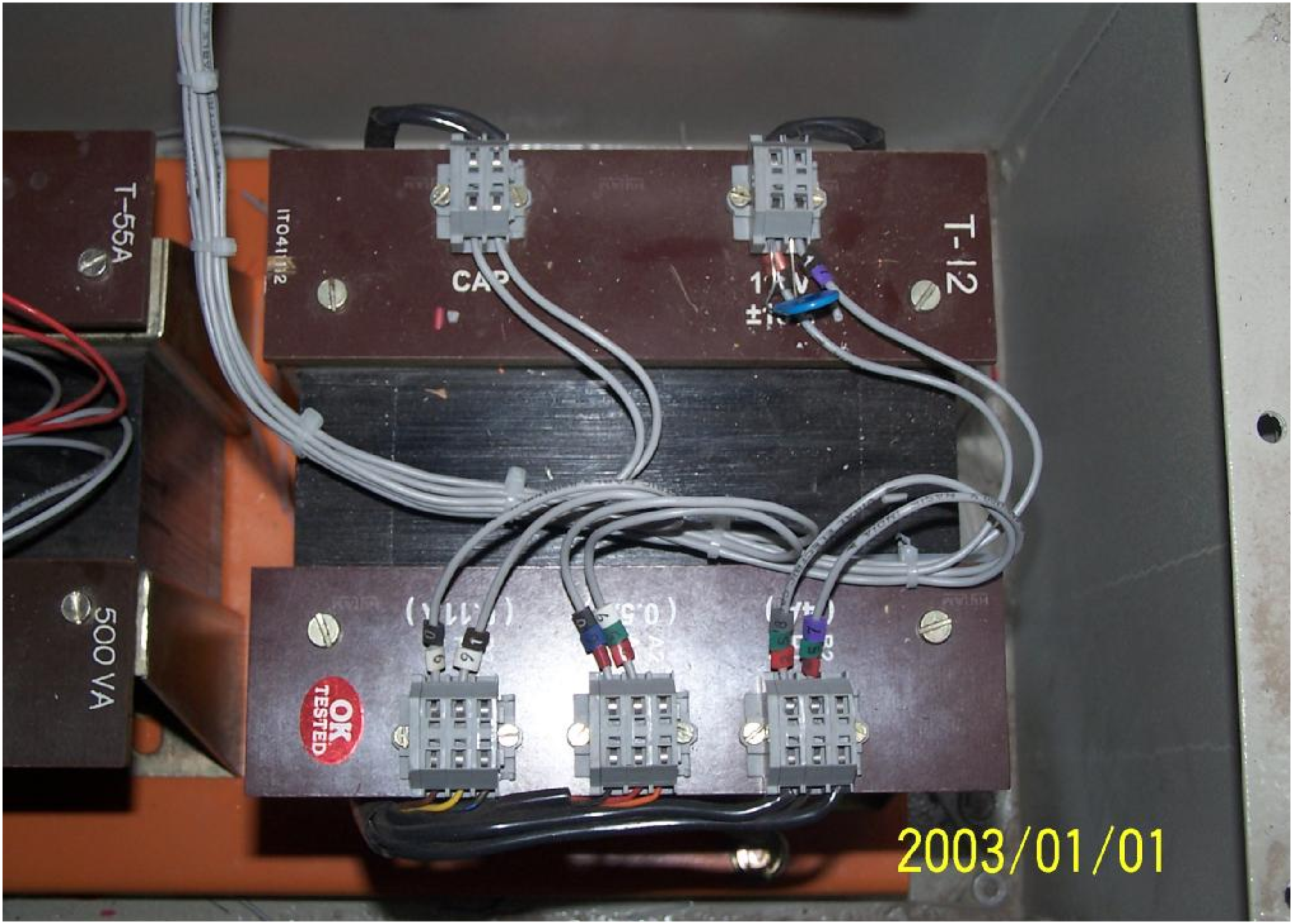
Various transformer used in the generator panel for peculiar function is as below:

- T-55A and T-55B: It is a step down transformer which convert 415V to 110 V.These 110V is are used to operate Start Contactor, Fed to start transformer,and fed to CVT.
- T-12 (CVT) : These is a constant voltage transformer used to supply constant voltage to ET-PS card. From 110 V it converts to 20V, 24V & 24V.
- T-17(Start Transformer): It is step up transformer feeding the voltage to charge the start capacitor. Primary has 110V and secondary from 400 to 700V .
- EMI –36(Feedback Transformer) : A step down transformer , which step down MF voltage from 3000V/3600V sensed from load circuit to 100 V used in control circuit.



SEC-2
SEC-1
PRI
T-68
OK TESTED
2003/01/01


500 VA
500 VA
T-55A
2003/01/01

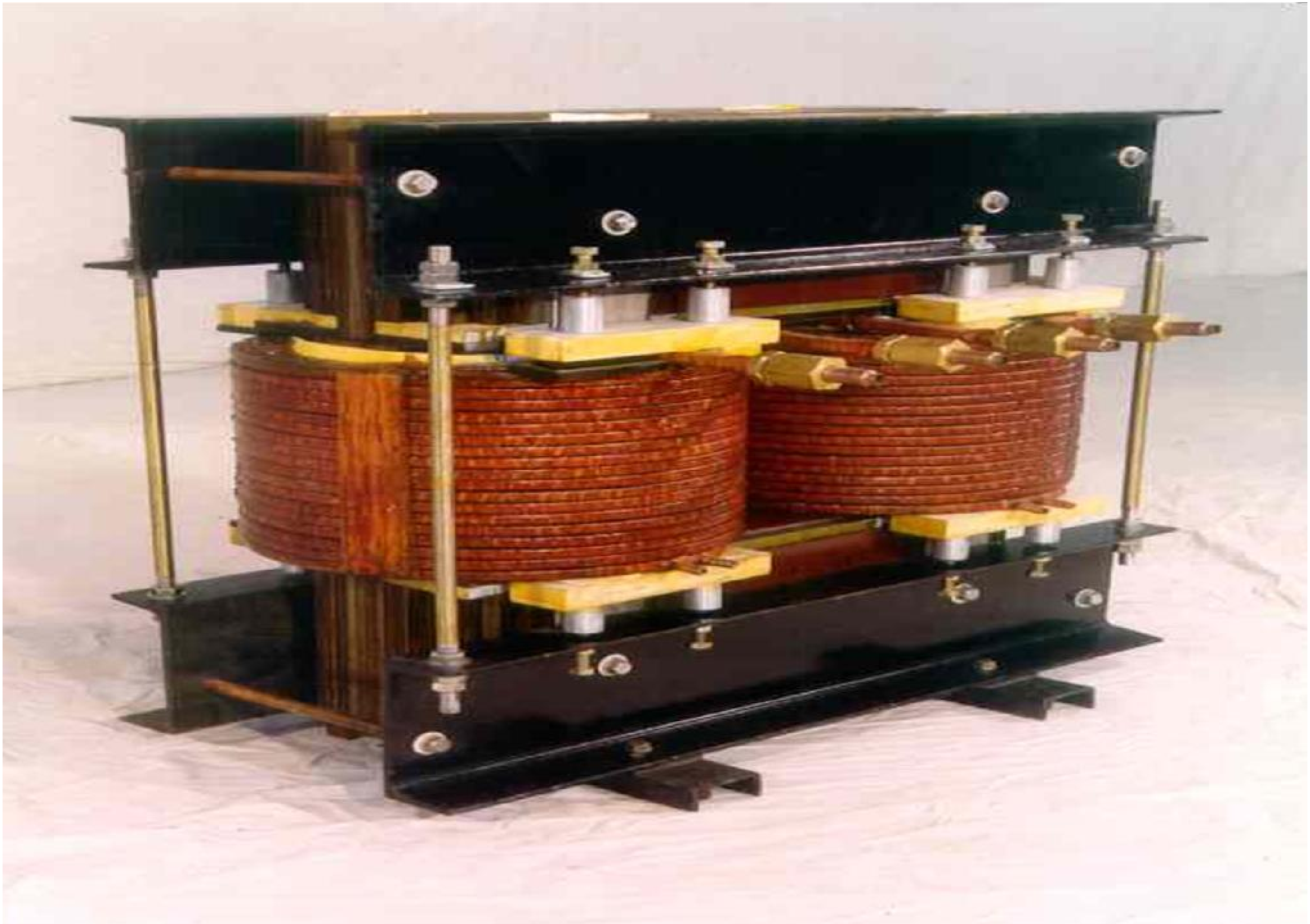

T-55A
500 VA
CAP
T-12
OK TESTED
2003/01/01

DC CHOKE

- Works on the principle of transformer, I.e primary and secondary are linked by magnetic flux. Transformer is a electro-magnetic device so here in DC , Choke acts only as resistance , which is in microohms.
- It is connected in series of Convertor output. One coil is connected to positive busbar and other to negative busbar.
- This is a large water- cooled Iron Cored Choke, which smoothens the output of the rectifier and provides a virtually constant current source for the Inverter.
- Coil of DC Choke acts as Inductor , so any change in current is resisted by this and therefore supply constant current to load circuit.

DC CHOKE ....cont

- At every half cycle of medium frequency we are firing other two pair of inverter thyristor. During switching time there is one such stage, when all inverter thyristor are in ON condition, hence short circuiting and due to this heavy current will flow , this short circuiting current is restricted to safe limit by DC Choke.
- Stacks of thin CRGO electrical steel are used as lamination packets. These steel sheets alloyed with silicon reduce the problem of ageing and improves the permeability and consequently reducing the magnetizing current and core losses.
- CRGO is made from ferrous base having maximum magnetizability I.e permitting a high induction. Both the side surface of CRGO sheet are coated with oxide coating .
- For reducing the eddy current losses thinner laminations are preferable.

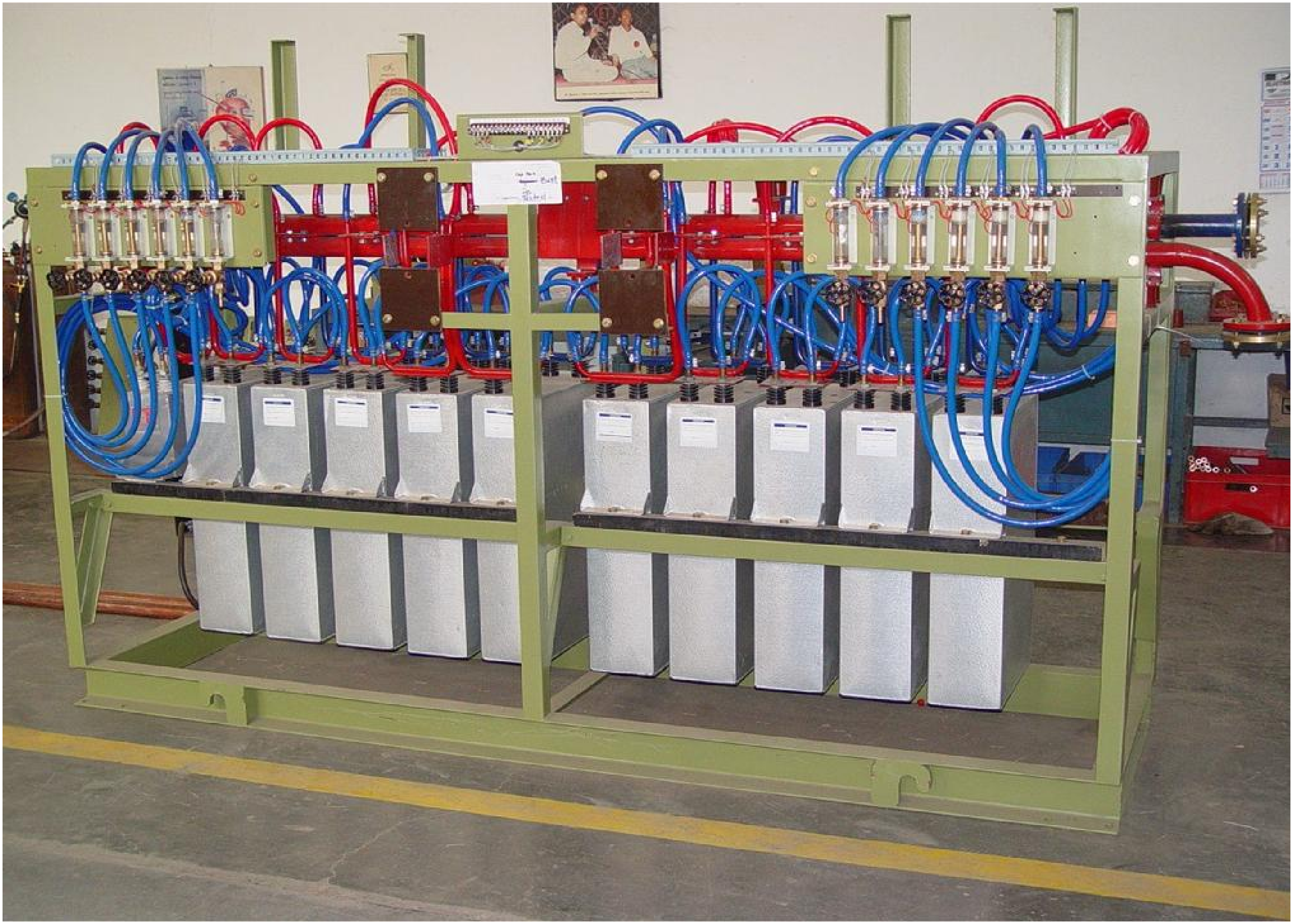


CHANGE OVER SWITCH

- This is offload switch used to alternately connect two furnace crucible with single power source.
- It works on CAM lever mechanism , can be operated manually or pneumatically. It is water cooled to allow better heat dissipation.
- Special damage control microswitch to trip the panel in case the switch is opened ON-Load.
- The current carrying capacity is from 2KA to 20KA.

CAPACITOR RACK

- Coil used for melting is highly inductive. Power factor of it alone is in range of 0.1. To bring the power factor near to unity compensating KVAR has to be added, and hence the capacitor bank.
- It also forms parallel resonance circuit with coil, providing the maximum current required by Coil.
- Cut-Off switch is provided to add or remove capacitor during melting process. Change over switch is mounted on this rack to select one crucible at a instant to deliver the power to coil.
- Water Flow switches is incorporated which display the DM water flow in water cooled Power capacitor, di/dt coil , cut-off switch & Change over switch. Incase any path is blocked due to any reason switch will operate and power generator panel will be switched off.



DEMINERALISE WATER CIRCULATION UNIT

- This unit has DM water reservoir, valve, water pump, strainer filter, heat exchanger, pressure gauge, FRP cylinder (mix bed resin cartridge)
- DM Water is sucked by pump from reservoir and delivered to filter and then passes through heat exchanger where this water is Cooled down by Raw water through exchanger plates.After passing through heat exchanger DM water is fed to generator inlet from where water is fed to different section of generator unit .
- Extracting heat from those units & finally return back to water reservoir.To avoid any scale formations in the internal cooling systems of Generator, cap rack, etc., due to presence of dissolved solids in the circulating water it is very much essential to use demineralized water having conductivity 10microsimens/cm or less than that.

DEMINERALISE WATER CIRCULATION UNIT ....cont

- Even though the DM water is being circulated in the cooling system, after some of the cycles, the conductivity of the demineralised water is gradually increasing due to the dissociation of ionizable metallic ions or ionizable non-metallic ions/traces. Therefore a fraction of the flow of circulating water is being passed through the mixed bed cartridge incorporated in the system & mixed regenerated ion exchanger resins filled up in the cartridge will lower down the conductivity of the circulating water by exchanging the undesirable ions with hydrogen (+)

  & hydroxide (-) ions fixed up in the resins by regeneration.

HYDRAULIC UNIT

- It consists of Electric motor coupled with Gear Pump, Return Check Valve, Pressure Release Valve, Direction Control Valve, Return Filter, Tank, Suction Filter and a pair of Hydraulic Cylinder-Piston combination.
- Electric motor coupled with Oil Pump, provides necessary pressure for tilting of furnace. This pressure is maintained with the help of Pressure Release Valve, which delivers extra oil in the tank thus maintaining required pressure.
- The Oil is passed through the Direction Control Valve, which controls the flow of oil. By controlling this valve the oil is passed to the tank in case the lifting is not required and in case lifting is needed the oil is passed to the Hydraulic Cylinder. Oil used is Enklo-68. It's characteristic is that it has High Viscosity, so it offers less friction and hence wear is less. The most important characteristics is that it does not allow formation of bubbles.

COIL CRADLE ASSEMBLY

- The helical water-cooled copper conductor coil surrounds a refractory crucible in the coreless induction-melting furnace.
- The coil is surrounded by a number of lamination packets made up of CRGO (Cold Rolled Grain Oriented) steel in order to provide easy path to the magnetic flux.
- The furnace inner assembly comprising the coil, lamination packets, top and bottom sections and cooling turns are bolted together with a final height setting to form a “Squirrel Cage”, which, if required, can be removed as a complete unit for repair.
- Coil cradle assembly is made of three parts viz; Bottom Ring, Coil & Top Ring.

BOTTOM RING

- The bottom section is made up of heavy mild steel shield with an aluminum sheath inside. This aluminum sheath provide easy magnetic flux path so that problem of bottom heating is avoided due to stray losses.
- On the inner surface it has castable cement refractory (K-54 cement with S.S. wires) on which Fire clay bricks are laid generally it is of three shapes and the gaps are completely filled with irlem Powder and solution (to avoid getting hardened the solution is made in small quantity).
- At the center of the bottom section, there is an insert made up of stainless steel called Antenna or Spider Assembly.
- It serves the purpose of Earthing and gives an indication whenever the lining is damaged and comes in contact with charge before charge making contact with coils and hence protects molten metal to have a direct contact with the furnace coil.


2003/01/01

FURNACE COIL

- Furnace coil are made of High conductivity, Water cooled , thick extruded rectangular section of electrolytic grade copper. This ensure extra efficiency , strength and rigidity.
- For maximizing the electrical efficiency of coil, correct spacing between the turns is maintain. Coil is coated with High voltage insulation paint , which provide maximum insulation between turn to turn, so that inter turn sparking is avoided.
- Coils are rigidly clamped both axially and radially to prevent motion that might result in chafing and damage to the insulation. For this studs are braze to individual turn , and this are bolted to FRP Support.
- Additional Stainless steel cooling turn at the top and bottom of the active coil is to provide uniform cooling which reduces the thermal stress on the lining. It also eventualy distribute temperature gradients through out the refractory to prevent overheating and extend refractory life.

TOP RING

- The top section is made up of heavy mild steel shield with an inner lining of castable cement refractory.
- In between this two there is a copper sheath also known as Faraday Ring, to provide path to the magnetic flux. Over Faraday Ring hollow copper pipes are brazed through which water flows in order to cool the apparatus.
- Inner lining of castable cement refractory is made up of mixture of K-54 cement and small pieces of S.S. wire. These two combination works as partial conducting material and hence gives a path too flux as it has small pieces of S.S. wire therefore there is no eddy current formation.

TILTING STRUCTURE

- Tilting structure is made from heavy channel of mild steel. The construction is so rugged that it can easily handle electromagnetic vibration created during melting cycle.
- It consist of Outer Base structure, which is fixed to foundation bolt for support. Inside is coil holding structure, which is connected to outer structure through Hinge . This inner structure having coil assembly fitted can be tilted with the support of two hydraulic arm.
- Twin hydraulic cylinders fed with hydraulic oil at pressure up to 150 kg/cm^2 .Viscosity of the hydraulic oil ranges from 100 to 300 centistokes. A pressure relief valve mounted on the hydraulic power pack adjusts necessary lifting pressure. To control the return speed, a throttle check valve is provided near the bottom of the cylinder.


BOX TYPE
FURNACE ASSEMBLY

SAFETY NORMS IN INDUCTION FURNACE

Following are Tripping incorporated in the Furnace system to fully protect it against any fault , which could lead to damage of Furnace component.

- Over Current
- Over Voltage
- Over Frequency
- Fuse Blown of Incoming Lines
- Door open
- Earth Leakage
- Water Conductivity
- Water Flow in generator, Coil and Outgoing Busbars.
- Water Pressure in Coil.
- Water temperature of Generator Panel
- Water temperature of Coil Water.

TROUBLE SHOOTING

Fault	Remedy
Ready lamp not glowing	Check for interlock failure and it's indication on panel.
MCCB ON LED not glowing even though the MCCB is ON.	Check Auxiliary NO contact of MCCB and it's functioning.
Fuse Blown	Check the fuses and its micro switch.
Generator water failure.	Check the floats of all switch , if all are lifted then check proximity switch.
Generator water temperature.	Heat Exchanger is choked up. Insufficient raw water flow rate. Check foot valve , pump & strainer.
Water Conductivity	Replace the resin of Deionized cartridge with new one.
Door Open	Ensure all door are closed and door switches working

Furnace is Not Switching ON , Trips in Over Current.

- **Probable reason behind it,**
  - Failure of power Components
  - No Feedback Signal
  - Converter , Inverter Firing pulse missing
  - Faulty Phase Sequence
  - Faulty Trip Setting , faulty PCB cards
  - Faulty MF Capacitor or its body touching earth
  - Faulty Supply voltage of +/- 15VDC, 26 VDC
  - Feed Back Fuses Open, Faulty Snubber circuit
  - Water Conductivity very High

Detail List

Furnace Switch ON but trips in Over Current

- **Probable reason behind it,**
  - Snubber circuit faulty.
  - Sharing Resistance Open
  - Phase current unbalanced
  - Faulty MFCT
  - Faulty Feed Back Transformer
  - Input Sine wave highly distorted
  - MF capacitor touching the frame.

Detail List

Furnace trips with out indication

- **Probable reason behind it,**
  - Loose connection in interlock circuit.
  - Earthing not proper.
  - Gate Cathode value of inverter thyristor.
  - Loose connection of wire in ET-133/2, ET-133/3, ET-77A & ET-77B.
  - Loose connection in ET-ITC & ET-SL.

Detail List

Failure of power component (Fuses, Thyristor & Capacitor)

- **<u>Probable reason behind it,</u>**
  - Water Temp. setting to high
  - Stretch TOT above 300 micro sec.
  - Improper Voltage Sharing.
  - Heavy scaling in cooling path of thyristor.
  - Giving full power to a cold charge I.e former
  - R35 in ET-94 card is not present.

<u>Detail List</u>

Starting Contactor Is Not Releasing In Test Mode

- **<u>Probable reason behind it,</u>**
  - Fault in Crucible.
  - Priming Current setting not proper.
  - Auxilliary contactor off.
  - Faulty Traic of ET-SC card.
  - Test / Run switch in run mode.
  - Control supply not proper.

Detail List

Furnace Not Picking Full Power

- **<u>Probable reason behind it,</u>**
  - Former size not proper.
  - Improper Voltage Limit , Current limits Stretched TOT , Input voltage or KW meter callibration.
  - Bottom height of Ramming Mass.
  - Loose Scrap , empty crucible.
  - Low input voltage.

  - <u>Detail List</u>

Average Power Factor of Furnace is Low I.e below 0.85

- **<u>Probable reason behind it,</u>**
  - Improper Voltage and current limit.
  - Furnace operating at low power long time.
  - Low PF of LT load
  - More patching or Lining heats in a month.
  - Faulty KWH or KVARH meter.
  - DC voltage at DC busbar with multimeter at voltage limit.

Detail List

Earth Leakage Trip

- **Probable reason behind it,**
  - Lining condition faulty.
  - Check any scrap particle between coil and lamination packet
  - Check moisture in lining. Check resistance between M.F. busbar and earth. It should be more than 500 ohms.
  - Faulty insulator of M.F. busbar.
  - Insufficient insulation between coil and top or bottom .
  - Faulty earth leakage C.T. Replace with new ELU CT .
  - Faulty earth leakage PCB